

# Notch and Catch: Dynamic Inefficiency in the Corporate Income Tax Schedule

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## Abstract

Notched corporate income tax schedules are prevalent worldwide, yet their effects on dynamic allocative efficiency remain understudied. Leveraging detailed administrative employer-employee linked tax return data from Taiwan, alongside a significant reform shifting the exemption threshold, we analyze the dynamic consequences of corporate tax notches. We document substantial bunching in the corporate income distribution, as firms strategically respond by reducing real output and shifting profits toward labor salaries to position themselves just below the threshold. Moreover, we identify a “notch trap”: firms that bunch below the exemption threshold become persistently constrained, experiencing 20% lower average revenue growth and an 8.1% higher exit rate relative to firms just above the threshold.

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# 1 Introduction

Corporate tax policy has changed substantially in recent decades, as governments have reduced statutory rates, narrowed tax bases, and introduced preferential regimes for smaller firms. These reforms are often motivated by the view that lower corporate taxes can stimulate investment, expand production, and support economic growth. At the same time, critics emphasize that such policies may generate limited real activity while imposing substantial fiscal costs through foregone tax revenue. One important example is the tax notch, under which tax liability changes discretely once taxable income crosses a threshold. Such notches are widely used in corporate income tax systems and are often justified as a way to support small and medium-sized firms while maintaining revenue from larger firms. Yet their discontinuous structure creates strong incentives for firms to remain below the threshold. Existing work has primarily examined static responses near notches, such as bunching and income misreporting. Less is known about whether these incentives also alter firms' decisions over time. By discouraging firms from crossing the threshold, tax notches may affect revenue growth, firm survival, and aggregate productivity.

This paper studies these dynamic consequences in the context of Taiwan's 2010 corporate income tax reform. The reform raised the taxable income exemption threshold from 50 thousand NTD to 120 thousand NTD, substantially expanding the set of small firms exempt from corporate income tax. We combine administrative records on corporate income tax filings, value-added tax filings, personal income tax records, and household registration data. The combination of annual corporate tax records and third-party-reported value-added tax records allows us to separate real responses from reporting responses. Employer-employee records linked to kinship information further allow us to document misreporting channels within family networks. Finally, the extended panel structure enables us to study how firms adjust not only around the threshold in the short run, but also along dynamic margins such as growth and survival.

Our empirical analysis proceeds in two steps. In the first step, we document how firms respond to tax notches. We find that firms exhibit pronounced bunching patterns in their taxable income distribution at the exemption thresholds before the 2010 corporate income tax reform. After the reform, firms that were previously bunched below the previous exemption threshold (50 thousand NTD) inflated their taxable incomes to bunch below the new exemption threshold (120 thousand NTD). This change is immediate, and common across different industries.

How do firms achieve bunching? We demonstrate two primary channels underlying the bunch-

ing behavior: real responses by shrinking production size, and misreporting by shifting profits as labor expenditures. To separate these two channels, we analyze changes in the average shares of intermediate input costs and labor salaries in gross revenues across taxable income bins before and after the reform. We rely on both cross-sectional variations induced by the exemption thresholds and the changes in the exemption thresholds by the 2010 corporate income tax reform to examine firm responses qualitatively.

Specifically, we find that firms with taxable incomes just below the exemption threshold have discontinuously smaller intermediate input per unit of gross revenues, both before and after the reform. This pattern suggests that firms below the exemption threshold tend to scale down production by utilizing less intermediate input. On the contrary, firms with taxable incomes just below the exemption threshold have discontinuously larger labor salary per unit of gross revenues. Moreover, exploiting rich information in our employee-employer linked records with family registration, we find that firms mainly shift corporate profits as personal incomes paid to relatives of firm owners. This suggests that firms exploit relatively lower personal income tax rates to avoid paying higher corporate income taxes and reduce the overall tax burdens. Through a battery of robustness checks, we rule out alternative channels for firm bunching such as revenue under-reporting and ownership restructuring.

We next examine the dynamic consequences of firm bunching. By comparing firms with taxable income immediately below and above the exemption thresholds, we find that those bunching firms tend to have a lower revenue growth rate and a higher exit rate. Those firms which bunch below the exemption thresholds have shrunk their operation to be lower than the optimal level in the absence of tax incentives. They require extra favorable growth opportunities to justify growing beyond the exemption thresholds, but will become more vulnerable to adverse cost shocks and are more likely to be eliminated from the market. Therefore, we uncover evidence of a “bunching trap” created by tax notches, where firms become persistently stagnant and eventually exit.

In the second step of our analysis, to further boost the confidence in empirical findings, we complement the bunching-based analyses with a regression framework, which relies on an exogenous shift in the relative distance between firms’ taxable incomes and the exemption threshold due to the 2010 corporate income tax reform. We include a rich set of fixed effects and firm size controls in the regression to account for other potential confounding factors. Our identification relies on the assumption that, conditional on firms’ taxable income, the variation in relative distance to the exemption threshold arises exogenously from policy changes and shifts firms’ incentives to bunch.

We find that conditional on firm size, industry-specific macro shocks, and other firm-level control variables, firms whose taxable incomes falls below the exemption threshold due to the 2010 tax reform tend to reduce intermediate input share in gross revenue by 2% and increase the salary shares in gross revenue by 7%. Their growth rate is 2.1 percentage points lower, which is approximately 20% of the sample mean. Moreover, their exit rate is 0.7 percentage points higher and approximately 8.1% of the sample mean. These empirical findings are fully consistent with our bunching-based analyses.

To quantify the welfare consequences from the tax notches, evaluate winners and losers from the reform, and explore alternative designs of the tax schedule, we develop a parsimonious static model of firm responses to a notched tax schedule. Firms face heterogeneous misreporting costs and choose between operating unconstrained or bunching at the threshold using a combination of real output reduction and reported labor inflation. The model partitions firms into three regimes: below-threshold firms, bunching firms with either interior or saturated responses, and non-bunching firms. It then uses a small set of population-average parameters to link static discontinuities in input shares to dynamic outcomes, including lower revenue growth and higher exit among bunching firms. We calibrate these parameters from the empirical moments documented in our reduced-form analysis. Using the calibrated model, we simulate the 2010 reform and decompose its welfare consequences. We find that the reform substantially reduces lifetime deadweight loss, with most of the gain accruing through the dynamic exit margin rather than through the static reduction in bunching distortion. Decomposing the reform into a structural change (threshold and schedule simplification) and a mechanical rate change (25% to 17%), we find that the structural change disproportionately benefits medium-productivity firms, while the rate change is a more uniform transfer to firms paying tax in both regimes.

Our paper contributes to the following two strands of literature. First, this paper contributes to a substantial body of work analyzing firms' behavioral responses to the notched corporate income tax schedule. The literature generally focuses on estimating the implied elasticity of corporate taxable income from static bunching patterns ([Auerbach, Hines Jr, and Slemrod, 2007](#); [Dwenger and Steiner, 2012](#); [Bachas and Soto, 2021](#)) or differentiating specific real and reporting responses in a static setting ([Devereux, Liu, and Loretz, 2014](#); [Coles, Patel, Seegert, and Smith, 2022](#); [Lobel, Scot, and Zúniga, 2024](#)).<sup>1</sup> Our paper provides firm-level evidence on firms' dynamic growth, entry,

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<sup>1</sup>[Devereux, Liu, and Loretz \(2014\)](#) explore the option for owner-managers to over-report deductible expenses. [Bachas and Soto \(2021\)](#) decompose the elasticity of corporate taxable income into elasticity of revenue and deduction costs, using a unique corporate tax schedule based on gross revenue in Costa Rica.

exit, and productivity, in addition to their static responses, and distinguishes between real and reporting strategies. Moreover, there is a growing literature on firms’ dynamic responses to size-based thresholds ([Liu, Lockwood, and Tam, 2022](#); [Choudhary and Gupta, 2024](#); [Garbinti, Goupille-Lebret, Muñoz, Stantcheva, and Zucman, 2023](#); [Muthitacharoen, Wanichthaworn, and Burong, 2021](#)), which mainly focuses on value-added tax thresholds defined in terms of gross revenue. We extend this by documenting firms’ dynamic responses to the notched corporate income tax schedule.

Second, our paper relates to the broader literature examining the effects of regulations and reforms on market efficiency and dynamics, including R&D incentives ([Chen, Liu, Suárez Serrato, and Xu, 2021](#)) and labor regulations ([Braguinsky, Branstetter, and Regateiro, 2011](#); [Gourio and Roys, 2014](#); [Garicano, Lelarge, and Van Reenen, 2016](#); [Amirapu and Gechter, 2020](#)). Our policy simulations aim to contribute to this understanding by analyzing the aggregate equity and efficiency implications of corporate income tax notches ([Swonder and Vergara, 2025](#)). Additionally, our study offers new perspective on “missing middle” size distribution patterns ([Dharmapala, Slemrod, and Wilson, 2011](#)) and the origins of small and medium-sized enterprises (SMEs) through the lens of tax policy. For example, [Aw, Chung, and Roberts \(2003\)](#) empirically highlights that Taiwanese manufacturers experience a higher turnover rate compared to those in South Korea, underscoring unique dynamics in SME formation.

The remainder of this paper is structured as follows. Section 2 describes the overall corporate taxation in Taiwan and the corporate income tax reform in 2010. Section 3 introduces the data used for our empirical analysis. Section 4 presents bunching-based analyses of how firms respond to tax notches, followed by the regression analysis exploiting the 2010 reform. Section 5 develops a simple model of firms with a notched corporate income tax schedule. Section 6 calibrates the model to the empirical moments documented in Section 4. Section 7 uses the calibrated model to simulate the 2010 reform, decompose its welfare consequences, and characterize its distributional incidence. Finally, Section 8 concludes.

## 2 Policy Background

### 2.1 Corporate Taxation in Taiwan

Corporate income tax has been the largest source of gross tax revenues in Taiwan. Over the period 2000–2024, Taiwan’s annual gross tax revenue averaged 2.1 trillion NTD, of which corporate in-

come tax contributed 23%, personal income tax contributed 20%, and value-added tax contributed 16% (Appendix Figure A.1). All profit-seeking enterprises operating within Taiwan are potentially subject to the corporate income tax.<sup>2</sup> Corporate income tax is assessed annually on firms' gross profits, which are calculated as total revenues net of deductible costs, including expenditures on labor, materials, capital, and other qualifying business expenses. The standard corporate income tax schedule is progressive with notches: firms with annual taxable income below the threshold face a zero statutory tax rate; Once the taxable income exceeds the threshold, the statutory tax rate is applied to the entire taxable income, resulting in a discrete jump in total tax liability.

All corporations are required to file corporate income tax returns annually before the end of May for the preceding fiscal year, even if they are temporarily closed or undergo negative taxable income.<sup>3</sup> To ensure proper filing and compliance with corporate income tax regulations, Taiwan's tax authority conducts random audits and onsite inspections, particularly to detect unregistered firms, late filings, or delayed payments. The consequences of such violations can be severe. Businesses operating failing to submit timely annual tax reports may face fines, criminal charges, and civil liabilities. Non-compliance is monitored by the Ministry of Finance and may result in further financial penalties or criminal prosecution.<sup>4</sup>

Another important tax faced by profit-seeking enterprises is the value-added tax (VAT). Under the VAT system, a standard rate of 5% is levied on the value added at each stage of production and distribution, calculated as the difference between a firm's revenue and deductible input purchases. Firms are required to record and file transaction-level VAT returns every two months, which enables tax authorities to trace detailed business-to-business transactions including materials and capital input purchases. It ensures firms to accurately report and validate their sales and purchases to receive VAT credits.<sup>5</sup>

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<sup>2</sup>Sole proprietorship and partnership only file the corporate income tax but are not requirement to any payment, as their income are subject to the personal income tax at the individual shareholder level. Our paper exclude them from all analyses.

<sup>3</sup>Firms which meet eligibility criteria based on their gross revenues and profits can opt for simplified filing methods called the Expanded Paper Review. Due to a different tax schedule as well as low adoption rate, we exclude firms under the Expanded Paper Review from the analyses.

<sup>4</sup>Late payments are subject to a 1% penalty for every three days of delay, up to a maximum of 30 days. Beyond that period, daily interest is charged according to the postal savings one-year fixed deposit rate. Please refer to the official documents from Taiwan Ministry of Finance accessed at <https://www.etax.nat.gov.tw/etwmain/tax-info/understanding/tax-q-and-a/national/profit-seeking-enterprise-income-tax/penalty/2J251pa>. Under the Article 110 of the Income Tax Act accessed at <https://law.moj.gov.tw/ENG/LawClass/LawAll.aspx?pcode=G0340003>, the penalty for corporate income tax evasion can be up to three times the amount of tax evaded.

<sup>5</sup>VAT has exemption rules as well. Specifically, firms with monthly sales below 80 thousand NTD are exempt from VAT and are not included in the administrative dataset. Their sales are instead estimated by the tax authority and

## 2.2 The 2010 Corporate Income Tax Reform

Taiwan tax authority implemented one of the largest income tax reform in 2010, aimed at boosting the economy and alleviating the tax burdens on small and medium-sized enterprises. Small and medium-sized enterprises have been the backbone of the Taiwanese economy, accounting for 98% of all registered businesses, 30% of total sales, and 78% of total employment in 2010.<sup>6</sup> The Income Tax Act Amendment, proposed by the Ministry of Finance, was approved by the cabinet on March 5, 2009, and officially enacted in 2010.<sup>7</sup>

The reform introduced a reduction in marginal statutory tax rates and an increase in the exemption threshold, as illustrated in Figure 1. Before 2010, the corporate income tax schedule consisted of three tax brackets, with a lower threshold of 50 thousand NTD (approximately 1,600 USD). Firms earning below this threshold were exempt from taxation. However, once taxable income surpassed 50 thousand NTD, a statutory tax rate of 15% applied, increasing further to 25% beyond 100 thousand NTD (approximately 40 thousand USD).<sup>8</sup> The 2010 reform simplified the tax structure by reducing the number of tax brackets from three to two and raising the tax threshold to 120 thousand NTD (approximately 3,900 USD).<sup>9</sup> Firms with taxable income below this threshold remained exempt from corporate income tax. Above the threshold, a uniform statutory tax rate of 17% applied, replacing the prior tiered structure.

The reform was presented as a pro-growth policy, especially for small and medium-sized enterprises. The new statutory corporate tax rate of 17% was lower than that of mainland China (25%) and South Korea (22%), and closely aligned with Singapore (17%) and Hong Kong (16.5%).<sup>10</sup> However, the reform also sparked public debate. Critics voiced concerns over the potential loss of government revenue, with estimates of annual reductions ranging from 30 billion to 50 billion NTD. They also questioned whether the projected economic gains would be sufficient to offset the

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subject to a 1% tax rate. Firms with monthly sales between 80 thousand and 200 thousand NTD, certain restaurants, and financial institutions are also exempt from VAT and subject to the aforementioned simplified rules.

<sup>6</sup>The Small and medium-sized enterprises are defined as firms with annual revenues below 80 million NTD (or numbers of employees below 200) in manufacturing, construction, and mining sectors, and those with annual revenues below 100 million NTD (or numbers of employees below 100) in other industries. Source: White Paper on Small and Medium Enterprises in Taiwan accessed at <https://www.sme.gov.tw/article-tw-2853-13128>.

<sup>7</sup>More detailed information can be accessed at <https://www.ey.gov.tw/Page/9277F759E41CCD91/cfe7d369-e943-4a2a-8db8-cedc19384885>.

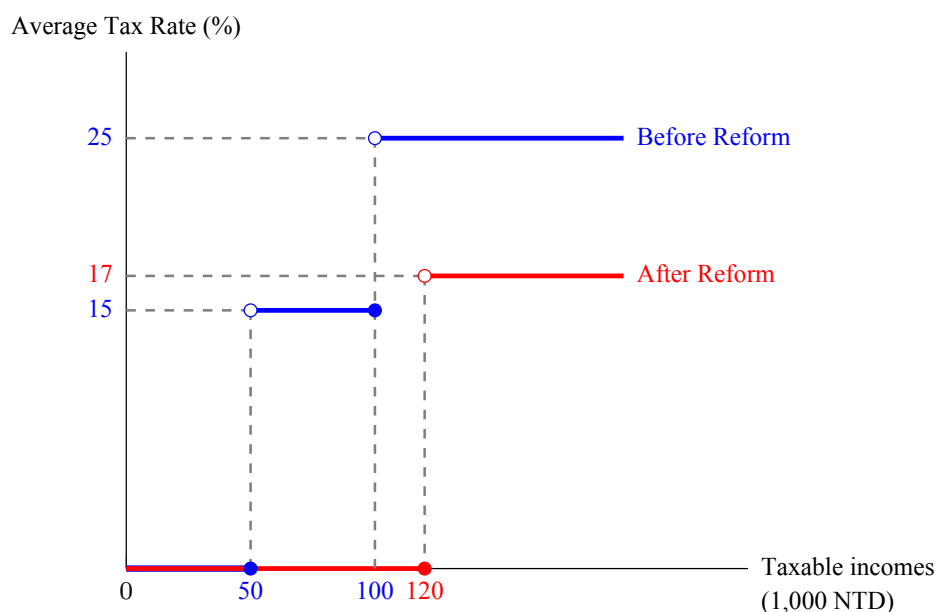
<sup>8</sup>To ease the tax burden on small enterprises, a buffer range was implemented at the first threshold, capping tax liability at 50% of the excess income until the average tax rate reached 15%.

<sup>9</sup>To ensure a gradual transition for firms near the threshold, the government maintained a buffer range, capping corporate tax liability at 50% of the amount exceeding the threshold until the average tax rate reached 17%.

<sup>10</sup>Please see <https://www.lawtw.com/archives/388050>.

fiscal cost of the reform.

**Figure 1: Corporate Income Tax Schedule**



*Notes:* This figure shows the corporate income tax schedule in Taiwan. The blue curve denotes the tax schedule before 2010, and the red curve denotes the tax schedule in 2010 and after.

### 3 Data

Our empirical analysis is primarily based on an employee-employer-linked administrative tax records maintained by the Ministry of Finance in Taiwan from 2004 to 2018. This data set covers corporate income tax and value-added tax records, personal income tax records, as well as household registration records. The universal firm coverage, long time series, and rich characteristics allow us to examine how firms respond to corporate income tax reform through both reporting behavior and real production adjustments.

The corporate income tax data provide comprehensive balanced-sheet information, including gross revenues, gross asset (fixed and liquid, including deposit), and a complete list of costs as well as their breakdown items. The value-added tax data provide information on business-to-business transactions and the intermediate input and capital purchases. Moreover, the tax records also cover the basic firm information including ownership, legal entity type, industry, establishment date, and location. Exploiting the panel data structure, we further construct firm revenue growth rate and firm

exit. We define firm exit as those disappearing from consecutive tax records. Given the mandatory nature of tax payment, firms that are missing in tax records over consecutive years have likely shut down.

Moreover, the personal income tax data provides comprehensive information on employee salaries. We use firm identifier to link the personal income tax records with the corporate income tax data and the value-added tax data, which yields comprehensive employer-employee linked records. We combine these data sets with household registration records through personal identification numbers, which help identify the kinship relationship between employees and firm owners.

We impose several restrictions to obtain our final sample. First, we keep firms that only subject to standard corporate income tax schedule, but exclude sole proprietorship and partnerships whose profits are subject to the personal income tax at the individual shareholder level.<sup>11</sup> Second, we exclude firms with negative taxable incomes, zero reported revenues, or missing employment information, which account for approximately 3% observation. Third, we restrict our sample to firms with taxable incomes between 0 and 300 thousand NTD (around 9,300 USD), covering the major corporate income tax threshold which increased from 50 thousand NTD to 120 thousand NTD.

Table 1 presents summary statistics for firm-level characteristics before and after the 2010 corporate income tax reform. We note four key data patterns. First, despite small taxable incomes, firms in our sample have sizable gross revenue averaged above 15 million NTD. Second, intermediate inputs are a dominant component of total cost, accounting for 70–72% of revenue, and labor salaries account for approximately 16% of gross revenue. Third, entry and exit rates are relatively stable across the two periods, with entry rates declining slightly after 2010, suggesting a relative stable yet active market turnover. Fourth, family-run firms are prevalent in our sample, as 70%-80% of firms have shareholders who are immediate family members (spouses, parents, siblings, and children). In addition, family members make up approximately 20% of the workforce, reflecting their significant roles in firm operations.

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<sup>11</sup>We also exclude firms with special filling method such as the Expanded Paper Review and tiny firms that are exempt from the VAT as discussed in Section 2.

**Table 1: Summary Statistics of Main Sample**

|                                       | 2004-2009 |      | 2010-2017 |      |
|---------------------------------------|-----------|------|-----------|------|
|                                       | Mean      | S.D. | Mean      | S.D. |
|                                       | (1)       | (2)  | (3)       | (4)  |
| <i>Panel A: Revenue and Costs</i>     |           |      |           |      |
| Gross revenue (million NTD)           | 20.0      | 23.4 | 16.9      | 21.3 |
| Intermediate inputs (million NTD)     | 15.4      | 20.0 | 12.6      | 17.9 |
| Labor salaries (million NTD)          | 2.5       | 2.8  | 2.1       | 2.6  |
| non-family-member employees           | 2.0       | 2.7  | 1.7       | 2.4  |
| family-member employees               | 0.4       | 0.4  | 0.4       | 0.4  |
| Number of employees                   | 11.2      | 13.3 | 8.6       | 11.4 |
| non-family-member employees           | 9.8       | 13.2 | 7.4       | 11.4 |
| family-member employees               | 1.4       | 1.0  | 1.2       | 1.0  |
| Capital investment Cost (million NTD) | 0.3       | 2.2  | 0.4       | 3.6  |
| Taxable Profit (thousand NTD)         | 81        | 76   | 103       | 72   |
| <i>Panel B: Firm Dynamics</i>         |           |      |           |      |
| Revenue growth rate                   | 0.11      | 0.67 | 0.09      | 0.62 |
| Entry rate                            | 0.10      | 0.29 | 0.07      | 0.10 |
| Exit rate                             | 0.10      | 0.30 | 0.11      | 0.10 |
| Firm age                              | 11.3      | 8.8  | 14.8      | 10.6 |
| Share of family firms                 | 0.73      | 0.28 | 0.79      | 0.27 |
| Number of observations                | 255,539   |      | 332,068   |      |

*Notes:* This table shows summary statistics before and after the 2010 corporate income tax reform (pre-reform pooled 2004-2009 in columns (1) and (2), and post-reform pooled 2010-2017 in columns (3) and (4).

## 4 How Firms Respond to Tax Notches

### 4.1 Bunching below the Exemption Threshold

We begin by examining the distribution of corporate taxable income as shown in Figure 2. The figure plots the number of firms across taxable income bins (2,000 NTD), averaged over the years before the reform (2004–2009, in blue triangles) and after the reform (2010–2016, in red circles). Before the corporate income tax reform, a clear bunching pattern appears at the exemption threshold of 50 thousand NTD. The number of firms increases gradually as taxable income approaches the threshold and then drops sharply above it, from roughly 900 firms in each 2,000-NTD taxable income bin below the threshold to fewer than 300 just above. Since the pre-reform tax schedule

had been unchanged for over twenty years, the discontinuity in the distribution of corporate taxable incomes indicates strong and persistent incentives for firms to strategically manage their corporate taxable incomes to remain just below the exemption threshold and thereby avoid corporate income taxes.

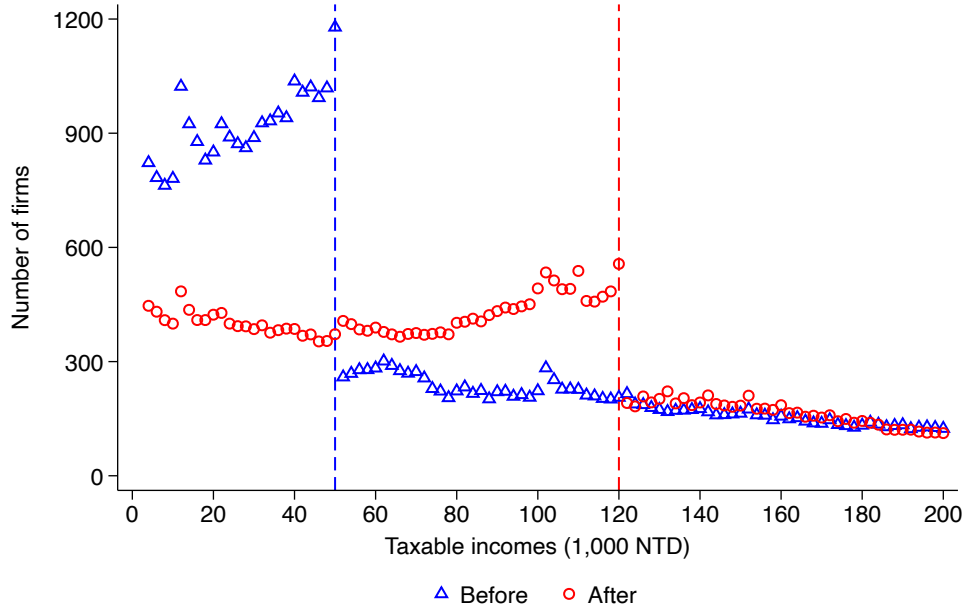
More importantly, the distribution of taxable income shifts noticeably in response to the change in the exemption threshold, indicating that firms adjusted their reported income around notches in the corporate tax schedule. After the reform, the discontinuity at the old threshold of 50 thousand NTD disappears entirely, while a new concentration of firms emerges just below the new exemption threshold of 120 thousand NTD. On average, there are approximately 420 firms in each 2,000-NTD taxable income bin below the new exemption threshold, with a modest uptick in counts in the bins closer to this threshold. However, there are approximately 165 firms per bin immediately above the exemption threshold. Among firms with taxable incomes below 120 thousand NTD, the share of firms below 50 thousand NTD decreased substantially from 73% to 39%. Those firms mainly increased their taxable incomes above 50 thousand NTD, with the share of firms reporting taxable incomes between 50 thousand NTD and 120 NTD increasing from 27% to 61%. In Appendix B, we quantify firm responses to the tax notches following the bunching literature ([Saez, 2010](#); [Kleven and Waseem, 2013](#); [Almunia and Lopez-Rodriguez, 2018](#)).

Beyond the pronounced bunching patterns at the exemption thresholds, a number of other noteworthy empirical patterns arise. First, we observe diffuse bunching with a noticeable concentration of firms just below the exemption threshold accompanied by a non-zero mass just above it, rather than a sharp spike. This pattern is consistent with the literature on bunching under frictions such as search costs or lumpy adjustment ([Anagol, Davids, Lockwood, and Ramadorai, 2024](#)).

Second, firms primarily respond to the exemption threshold but show little adjustment around the higher threshold in the tax schedule. For example, Figure 1 shows a tax notch at 100 thousand NTD prior to the reform where the average tax rate increased from 15% to 25%, but we find little evidence of bunching at this higher threshold. This limited response is consistent with the literature ([Kleven and Waseem, 2013](#)) and can be attributed to tax-rate misperception and other optimization frictions.

Third, firms' bunching responses are immediate and common across industries. The distribution of taxable income changed sharply between 2009 and 2010, as shown in Appendix Figure A.2, and remained roughly stable since 2010. The sudden increase in the number of firms reporting taxable incomes between 50 thousand and 120 thousand NTD in 2010 are mainly from those

**Figure 2:** Distribution of Corporate Taxable Incomes



*Notes:* This figure shows the average annual distribution of corporate taxable incomes before the reform (2004-2009, in blue triangles) and after (2010-2017, in red circles). The vertical axis denotes the average number of firms yearly in each taxable income bin. The horizontal axis represents the taxable income bins. The bins are \$20,000 wide, delimited such that no bin contains data both to the left and to the right of the relevant policy thresholds. For example, the bin 1 represents firms that have taxable income between 10 thousand NTD (included) and 12 thousand NTD (not included).

reporting between 30 thousand and 50 thousand NTD in 2009, according to Panel (a) of Appendix Figure A.3. Moreover, the bunching pattern remains consistent across sectors, since we observe similar bunching behaviors in wholesale, retail, manufacturing, and construction sectors, as shown in Appendix Figure A.4.

Finally, although the reform also reduced the statutory tax rate from 25% to 15% for firms with taxable incomes above 120 thousand NTD, the number of firms reporting taxable incomes between 120 thousand and 200 thousand NTD remains largely unchanged. Therefore, our analysis focuses on firms' behavioral responses to the exemption threshold and the dynamic implications of these responses. Firm responses to marginal tax rate changes above 120 thousand NTD are interesting in their own right (similar to evaluations on the Tax Cuts and Jobs Act in the US (Kennedy, Dobridge, Landefeld, and Mortenson, 2022)), but we leave the detailed empirical investigation to future work.

## 4.2 How Do Firms Bunch?

We next examine how firms adjust their taxable incomes to bunch below the exemption threshold. We demonstrate two primary channels underlying the bunching behavior: real responses by shrinking production size, and misreporting by shifting profits as labor expenditures. We distinguish these two channels by examining average shares of intermediate input, investment, and labor salary, relative to gross revenues. Figure 3 shows the average input cost shares in gross revenues with respect to the reported corporate incomes before and after the reform, respectively.

**Real responses** Panel (a) of Figure 3 plots the ratio of intermediate inputs and revenues across taxable income bins. We observe that firms with taxable incomes just below the exemption threshold have significantly smaller intermediate input per unit of gross revenues, both before and after the reform. Since the discontinuities in the ratios of intermediate inputs and revenues locate exactly at the old exemption threshold before the reform, and moves sharply with the shift of the new exemption threshold after the reform, we attribute the discontinuity to the corporate income tax notches and its changes. Other concurrent policies do not coincide with the corporate income tax reforms either in the notched structure or in the timing. Therefore, this pattern suggests that firms below the exemption threshold tend to scale down production by utilizing less intermediate input. Moreover, we plot the ratio of investment expenditures and revenues across taxable income bins in Panel (c) of Figure 3. It shows that the total investment exhibits a similar pattern, albeit less pronounced due to larger adjustment cost for capital.

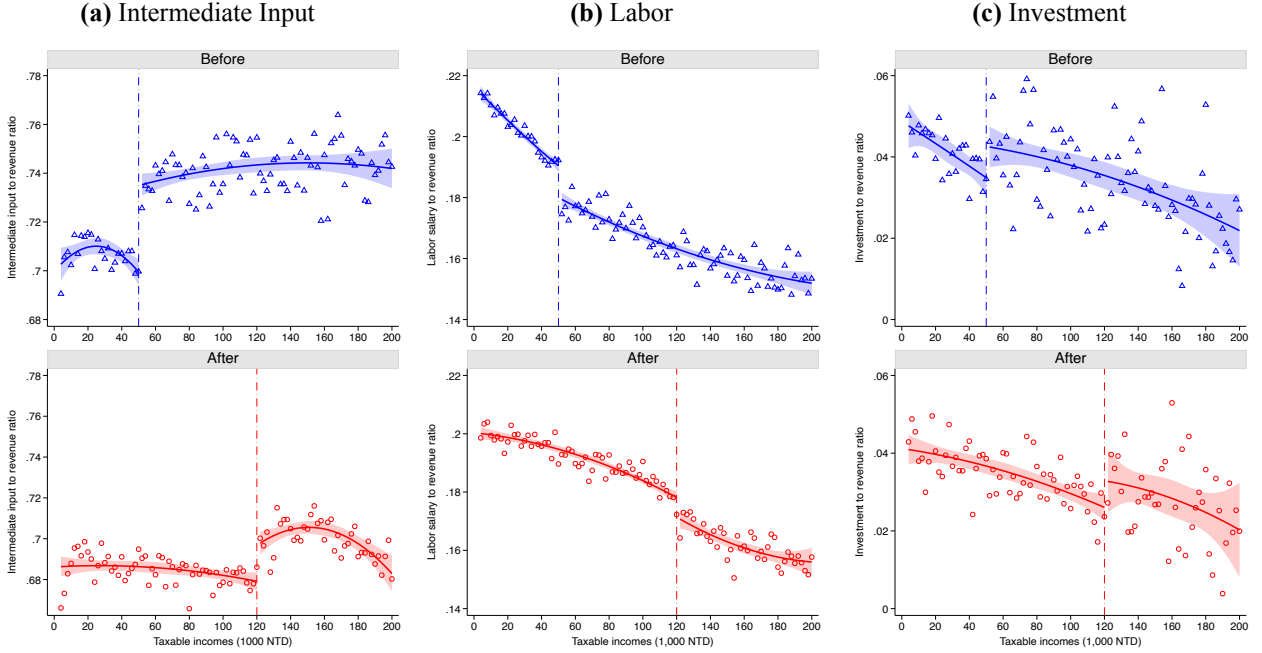
**Profit shifting** We plot the ratio of labor salaries and revenues across taxable income bins in Panel (b) of Figure 3. In contrast to the data patterns of intermediate inputs and capital investment, we find that firms with taxable incomes just below the exemption threshold have significantly higher labor salary per unit of gross revenues. This stark discontinuity suggests that firms strategically over-report labor costs to lower taxable income. We further examine which group of employees enjoy an increase in the salary, by exploiting rich information in our employee-employer linked records with family registration. We classify employees into two groups: those related to the firm's shareholder (family members) and those not related (non-family members).<sup>12</sup> As shown in Appendix Figure A.5, we find that firms mainly shift corporate profits as personal incomes paid to relatives of firm owners. Since the personal income tax (5%) is much lower than the corporate income tax above the exemption threshold (15% or 25% before the reform, 17% after the reform), the

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<sup>12</sup>We define relatives as the spouse, parents, siblings, or kids of the focal firm owner.

profit shifting behavior creates lucrative opportunities for firms to avoid paying corporate income taxes and lower the overall tax burdens.

**Figure 3: Input Revenue Ratios and Taxable Income**



*Notes:* This figure shows the average input revenue ratios with respect to corporate taxable income bins before the reform (2004–2009, upper panels with blue triangles and lines) and after (2010–2017, lower panels with red circles and lines). Panel (a) plots the ratios of intermediate input and gross revenue; Panel (b) plots the ratios of labor salary and gross revenue; Panel (c) plots the ratios of investment and gross revenue. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

**Other channels** We further investigate potential alternative mechanisms and find no evidence that firms strategically lower their reported taxable income by manipulating revenues. Average firm revenues across taxable-income bins increase smoothly with taxable incomes and show no visible discontinuity around the thresholds, as illustrated Panel (a) in Appendix Figure A.6. If firms were managing their taxable incomes primarily by adjusting reported revenues, we would expect to observe a sharp decline in total revenues just below the threshold. However, the data reveals no such pattern, suggesting that revenue manipulation is unlikely to be the main channel driving firms' bunching behavior.

To further address concerns about revenue under-reporting, we decompose firms' total revenue into business-to-business (B2B) and business-to-consumer (B2C) components and construct the

ratio of B2B revenues to total revenues. Under the value-added tax system, if firms under-report revenues, they would likely conceal B2C sales while leaving B2B transactions largely unchanged.<sup>13</sup> As shown in Panel (b) of Appendix Figure A.6, the B2B revenue share remains consistently around 70% across all taxable income bins, both before and after the reform, providing no evidence of systematic under-reporting through this channel.

Finally, we rule out the possibility that firms reduce their taxable income by splitting into multiple smaller entities. Using shareholder information, we identify “splitting” firms as those sharing at least one common shareholder with another firm and calculate the ratio of such firms to the total number of firms in each taxable-income bin. As shown in Appendix Figure A.7, this ratio remains stable at around 10–15% across all bins and over time, with no noticeable discontinuity at the threshold or change following the reform. These results suggest that ownership restructuring or firm splitting is unlikely to be the main driver of the empirical patterns we document.

### 4.3 Dynamic Consequences: Growth and Exit

We next examine the dynamic consequences of the bunching behavior. By comparing firms with taxable income immediately below and above the exemption thresholds, we document two related dynamic effects for those bunching firms. First, they tend to have a lower revenue growth rate. Second, they tend to have a higher exit rate.

**Slow-down of revenue growth** Panel (a) of Figure 4 plots the annual revenue growth rate with respect to taxable income bins. We define a firm’s revenue growth rate as the percentage changes in gross revenues between two adjacent years. We find that firms experience a slowdown in revenue growth if their taxable income is below the exemption threshold, and there is a sharp increase in the revenue growth rate for firms with taxable incomes above the exemption threshold. Similar to static decision, the discontinuity in the revenue growth rate moves sharply to the shift of the exemption threshold after the reform. We substantiate this pattern as a real response rather than a reporting behavior by examine the growth rate of revenues from business-to-business transactions and intermediate inputs, both of which are under scrutiny of the value-added tax system. As shown

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<sup>13</sup>The value-added tax system functions as a third-party reporting mechanism that enhances traceability for B2B transactions. Because B2B sales represent deductible inputs for buyer firms, they have an incentive to report these transactions to the tax authority. In absence of collusion between supplier and buyer firms, discrepancies in B2B sales are easily cross-checked, making them less prone to under-reporting. In contrast, B2C transactions are much less traceable, as final consumers lack incentives to retain or report receipts. See Kleven, Knudsen, Kreiner, Pedersen, and Saez (2011) and Carrillo, Pomeranz, and Singhal (2017) for discussions of transaction traceability and evasion.

in Appendix Figure A.8, the pattern is similar to what we observe in Figure 4.

**Increased exit rate** Panel (b) of Figure 4 shows the average exit rate across taxable income bins. We define firm exit as stopping submitting tax records. We find that firms whose taxable income is below the exemption threshold have a significantly higher exit rate, and there is a sharp drop in the exit rate for firms with taxable income above the exemption threshold. Using alternative definitions of firm exit, such as missing tax record in a certain year, or for three consecutive years, we find that the data patterns in Appendix Figure A.9 are similar to our baseline pattern.

Firms which bunch below the exemption thresholds have shrunk their operation to be lower than the optimal level in the absence of tax incentives. These bunching firms require extra favorable growth opportunities to justify growing beyond the exemption thresholds and incurring substantially higher tax liabilities. Therefore, they tend to remain small in size, but will become more vulnerable to adverse cost shocks and are more likely to be eliminated from the market. We thus uncover evidence of a “bunching trap” created by tax notches, where firms become persistently stagnant and eventually exit.

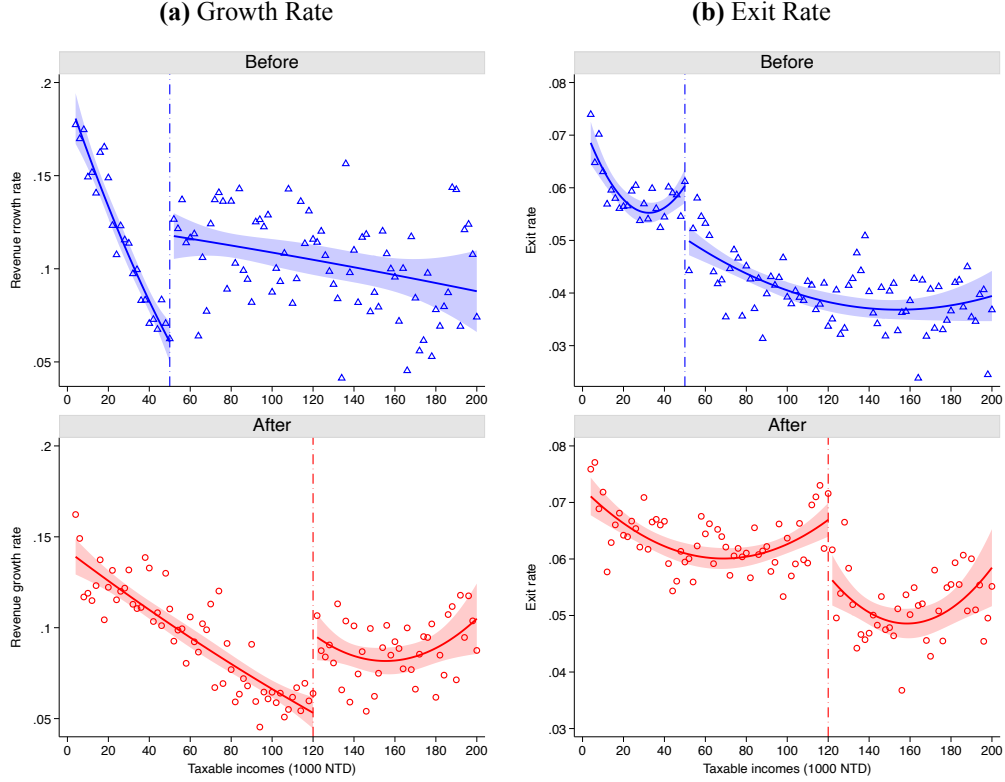
#### 4.4 Regression Analysis Based on the 2010 Tax Reform

**Empirical strategy** Our descriptive evidence rely on a cross-sectional variation induced by the exemption thresholds before and after the 2010 corporate income tax reform, respectively. We argue that our descriptive evidence documents the impacts of the tax notches, since we observe sharp discontinuities exactly at the exemption thresholds, which is unlikely to be an artifact from any other policy changes. To further boost the confidence in our empirical findings, we complement bunching-based analyses with a regression framework, which relies on a exogenous shift in the *relative distance* of firms’ taxable incomes and the exemption threshold due to the 2010 corporate income tax reform. We include a rich set of fixed effects in the regression model to control for other potential confounding factors. Our empirical model is as below.

$$y_{ijt} = \sum_{k=-5, k \neq 1}^5 \alpha_k \mathbb{1}(BinDistance_{it} = k) + \sum_{b=1}^{20} \beta_b \mathbb{1}(Bin_{it} = b) + \lambda_{jt} + X_{it} + \epsilon_{ijt} \quad (1)$$

We denote the outcome variable as  $y_{ijt}$ , including the average input cost shares in gross revenues, revenue growth rate, or the exit dummy for firm  $i$  in industry  $j$  and year  $t$ . We group

**Figure 4: Revenue Growth Rate, Exit Rate, and Taxable Income**



*Notes:* This figure shows the average revenue growth rates and exit rates with respect to corporate taxable income bins before the reform (2004-2009, upper panels with blue triangles and lines) and after (2010-2017, lower panels with red circles and lines). Panel (a) plots the average revenue growth rates; Panel (b) plots the exit rate where firm exit is defined as disappearing from the tax records. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

firms into taxable income bins of 10 thousand NTD,  $Bin_{it}$ . For example, the group of firms with  $Bin_{it} = 4$  include those with taxable incomes between 40 thousand (included) and 50 thousand NTD (not included). The variable  $BinDistance_{it}$  measures the gap between the exemption threshold and firm  $i$ 's taxable income such that

$$BinDistance_{it} = \begin{cases} 5 - Bin_{it}, & \text{if } t \leq 2009 \\ 12 - Bin_{it}, & \text{if } t > 2009. \end{cases}$$

A negative value of  $BinDistance_{it}$  indicates that the taxable income of firm  $i$  is above the exemption threshold. For example,  $BinDistance_{it} = -2$  for firms with  $Bin_{it} = 7$  (taxable income

between 70 thousand and 80 thousand NTD) in 2009, since the exemption threshold was 50 thousand NTD.  $BinDistance_{it} = 5$  for firms with  $Bin_{it} = 7$  (taxable income between 70 thousand and 80 thousand NTD) in 2010, because the exemption threshold increased to 120 thousand NTD.

We include a full set of dummies of  $BinDistance_{it}$  and  $Bin_{it}$ . Coefficients  $\alpha_k$  are of key interests as they capture how tax notches shape firm outcomes. The key identification assumption is that conditional on the firms' taxable income bins  $Bin_{it}$ , the variation in the relative distance to the exemption threshold  $BinDistance_{it}$  comes from the exogenous shift in the threshold from 50 thousand NTD to 120 thousand NTD by the 2010 corporate income tax reform. Consequently,  $\alpha_k$  compares how firm outcomes change with the distance to the exemption threshold, even holding their taxable income bins constant.

Moreover, we control for industry-by-year fixed effects,  $\lambda_{jt}$ , to account for common macroeconomic shocks over time and across industries, such as economic stimulus policies, trade shocks, inflation, or labor market conditions that could affect compensation for all firms. These fixed effects control for the impact of the 2008 global financial crisis and the subsequent economic recovery as well. The vector  $X_{it}$  includes firm-level covariates, including a firm's location, company type, age, and lagged assets, which capture potential differences in outcomes driven by firm characteristics unrelated to the relative distance between a firm's taxable income and the exemption threshold.

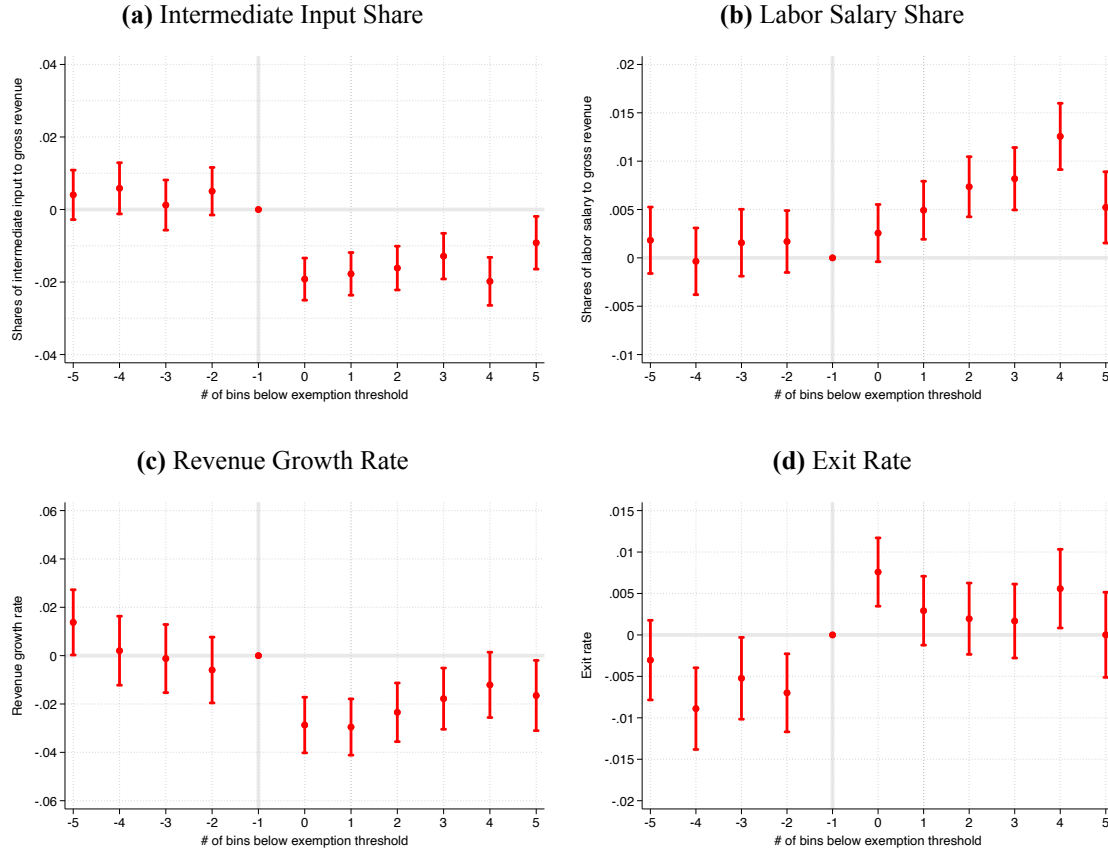
**Estimation results** We present results in Figure 5, which are consistent with the empirical patterns in Figures 3 and 4. Compared to firms above the exemption threshold ( $BinDistance_{it} < 0$ ), firms whose taxable incomes are below the exemption threshold tend to have a significantly lower intermediate input share in gross revenue, conditional on firm size, industry-specific macro shocks, and other firm-level confounding factors. The average intermediate input share in gross revenues is approximately 2% ( $= \frac{0.015}{0.75}$ ) lower for firms that are below the exemption threshold, which is sizable considering that the average taxable income to gross revenue ratio is 0.4%. Moreover, compared to firms above the exemption threshold ( $BinDistance_{it} < 0$ ), firms whose taxable incomes are below the exemption threshold tend to have a significantly higher labor salary share in gross revenue. The average labor salary share in gross revenues is approximately 6.7% ( $= \frac{0.01}{0.13}$ ) higher for firms that are below the exemption threshold. These results confirm the existence of both real responses and profit shifting for firms to avoid corporate income taxes.

Panels (c) and (d) further demonstrate the dynamic consequences of bunching due to tax notches. We find that firms whose taxable incomes are below the exemption threshold tend to have a much lower revenue growth rate compared with their counterparts above the threshold. The revenue

growth rate is 2.1 percentage points lower, which is approximately 20% of the sample mean. Moreover, firms below the exemption threshold also have a higher exit rate, which is 0.7 percentage points higher and approximately 8.1% of the sample mean.

The baseline empirical design relies on variation between firms within a firm size class. As a robustness check, we implement an alternative strategy that exploits within-firm variation in the same spirit of the traditional difference-in-differences design. We defer a detailed description to Appendix C. We examine the effects of lifting the exemption threshold from the 2010 reform on firms' intermediate input share and labor compensation shares, controlling for firm fixed effects. The results are fully consistent with our main findings. The same empirical framework is not well suited to study firms' dynamic responses, as we cannot control for firm fixed effects when analyzing exit decisions.

**Figure 5: Empirical Evidence Exploiting Changes in Distance to Threshold**



*Notes:* The figure displays the coefficient estimates for  $\alpha_k$  in equation (1) with 95% confidence intervals, capturing the treatment effect of tax notches on firm outcomes. The empirical model controls for firm corporate income bin dummies, industry-year fixed effects, and firm-level covariates, including location, company type, age, and lagged assets. The horizontal axis represents the relative distance of firms' taxable income from the exemption threshold, and positive number represents taxable income bins below the threshold. For example, the group with the relative distance 0 includes firms with taxable income between 40 thousand and 50 thousand NTD before the reform (when the exemption threshold was 50 thousand NTD) and firms with taxable income in the range of 110 thousand and 120 thousand NTD after the reform (when the exemption threshold was raised to 120 thousand NTD).

## 5 A Simple Model of Firm Responses to the Tax Notch

We develop a parsimonious static model of firm responses to a notched corporate income tax schedule. The model captures the two bunching margins documented in the empirical analysis: real output reduction and labor-cost over-reporting, and delivers closed-form implications for input shares, revenue growth, and exit. Firms differ in their misreporting curvature, which determines whether they use only the reporting margin or also reduce real activity. Adding exogenous productivity evolution and an outside-option shock allows us to map static distortions to the dynamic patterns we observe in the data. We then use the calibrated model to evaluate the welfare consequences of the 2010 reform and study the distributional incidence of the policy change.

### 5.1 Environment

We consider a firm that produces a homogeneous good using predetermined capital and two flexible inputs, labor and intermediate materials. Let  $z$  denote firm productivity and  $k$  denote capital. The contribution of productivity and predetermined capital is summarized by the effective productivity index  $A \equiv zk^\alpha$ . Output is produced according to

$$y = A \ell^\beta m^\gamma, \quad \theta \equiv \beta + \gamma < 1. \quad (2)$$

We normalize output, wage, and intermediate-input prices to one. The restriction  $\theta < 1$  implies decreasing returns to the flexible inputs, holding capital fixed.

For a given output level  $y$ , cost minimization yields the variable cost function

$$C(y; A) = c_0 A^{-1/\theta} y^{1/\theta}, \quad c_0 \equiv \theta \beta^{-\beta/\theta} \gamma^{-\gamma/\theta}. \quad (3)$$

True before-tax profit is therefore  $\pi(y; A) = y - C(y; A)$ . The unconstrained output choice and associated maximum profit are

$$y^*(A) = \left( \frac{\theta}{c_0} \right)^{\theta/(1-\theta)} A^{1/(1-\theta)}, \quad \pi^*(A) = (1 - \theta) y^*(A). \quad (4)$$

Absent the tax notch, revenue is isoelastic in productivity with elasticity  $1/(1 - \theta)$ . Cobb–Douglas cost-minimization implies constant true input shares of revenue at the unconstrained optimum:  $\ell/y = \beta$  and  $m/y = \gamma$ .

## 5.2 Tax Notch and Reporting

The corporate income tax schedule contains a single notch at reported profit  $\bar{\pi}$ :

$$T(\pi^r) = \begin{cases} 0, & \pi^r \leq \bar{\pi}, \\ \tau \pi^r, & \pi^r > \bar{\pi}. \end{cases} \quad (5)$$

Crossing the threshold creates a discrete increase in tax liability and gives firms with true profit slightly above  $\bar{\pi}$  an incentive to report at or below the notch.

Firms can reduce reported profit through two channels. First, they may reduce real output, which lowers true profit. Second, they may inflate reported labor expenses. Let  $\ell$  denote true labor expenditure and  $\ell^r \geq \ell$  denote reported labor expenditure. The reporting wedge is  $\Delta \equiv \ell^r - \ell \geq 0$ . Inflating reported labor by  $\Delta$  is costly:

$$g(\Delta; \psi_i) = \frac{\psi_i}{2} \Delta^2, \quad (6)$$

where  $\psi_i > 0$  is firm-specific. We assume  $\psi_i$  is drawn independently from a distribution  $H$  on  $(0, \infty)$ , independent of productivity. Heterogeneity in  $\psi_i$  allows firms to differ in the relative cost of using the reporting margin and is necessary to simultaneously match the magnitude of the reporting wedge and the existence of real-output distortion documented in Section 4. Reported profit is

$$\pi^r = \pi(y; A) - \Delta. \quad (7)$$

We restrict the reporting technology to a notch-specific use: firms either report truthfully,  $\Delta = 0$ , or use the reporting margin to locate exactly at the notch,  $\pi^r = \bar{\pi}$ . This restriction focuses the model on bunching incentives rather than on general intensive-margin tax evasion above the threshold.

## 5.3 The Firm's Static Problem

The firm chooses between unconstrained operation and threshold bunching. The no-adjustment payoff is

$$\Pi^0(A) = \begin{cases} \pi^*(A), & \pi^*(A) \leq \bar{\pi}, \\ (1 - \tau)\pi^*(A), & \pi^*(A) > \bar{\pi}. \end{cases} \quad (8)$$

Alternatively, the firm may choose output and reporting so that reported profit equals the threshold. The bunching payoff is

$$\Pi^B(A, \psi) = \max_{y \leq y^*(A), \pi(y; A) \geq \bar{\pi}} \{\pi(y; A) - g(\pi(y; A) - \bar{\pi}; \psi)\}. \quad (9)$$

The overall payoff is  $\Pi(A, \psi) = \max\{\Pi^0(A), \Pi^B(A, \psi)\}$ .

**The bunching trade-off.** Conditional on bunching at the notch,  $\pi^r = \bar{\pi}$  requires  $\Delta = \pi(y; A) - \bar{\pi}$ . Substituting this expression into the firm's objective shows that the bunching problem can be written as the minimization of two costs:

$$TC(y; A, \psi) = \underbrace{\pi^*(A) - \pi(y; A)}_{\text{real output distortion}} + \underbrace{\frac{\psi}{2} (\pi(y; A) - \bar{\pi})^2}_{\text{misreporting cost}}. \quad (10)$$

The first term is the profit loss from reducing output below its unconstrained level. The second term is the cost of inflating reported labor expenses enough to place reported profit at the threshold. Differentiating with respect to output yields the first-order condition

$$\psi \Delta = 1, \quad \text{or equivalently} \quad \Delta^\dagger(\psi) = \frac{1}{\psi}. \quad (11)$$

The optimal wedge interpretation is straightforward. Reducing reported profit by one unit through the real margin costs one unit of forgone profit. Reducing reported profit through misreporting has marginal cost  $g'(\Delta; \psi) = \psi \Delta$ . The firm uses the reporting margin until its marginal cost equals the marginal cost of reducing real activity. Firms with lower  $\psi$  rely more heavily on reporting adjustments; firms with higher  $\psi$  reach the real-output margin sooner.

## 5.4 Three Regimes

The model partitions firms into three regions according to their unconstrained profit and their misreporting curvature. Let  $\pi^* = \pi^*(A)$  for brevity.

**Regime I: below the threshold.** Firms with  $\pi^*(A) \leq \bar{\pi}$  are naturally below the threshold. They choose unconstrained output, report truthfully, and pay no tax. These firms are not distorted by the notch.

**Regime B: bunching.** Firms with unconstrained profit moderately above the threshold bunch at  $\bar{\pi}$ . The optimal reporting wedge is

$$\Delta(\pi^*, \psi) = \min\{\pi^* - \bar{\pi}, 1/\psi\}. \quad (12)$$

When  $\pi^* - \bar{\pi} < 1/\psi$ , the firm can reach the notch using only the reporting margin and continues to produce at  $y^*(A)$  (*interior bunching*). When  $\pi^* - \bar{\pi} > 1/\psi$ , the reporting wedge reaches its firm-specific upper bound and the firm must also reduce output to  $\hat{y}(A, \psi) < y^*(A)$  satisfying  $\pi(\hat{y}; A) = \bar{\pi} + 1/\psi$  (*saturated bunching*). Net cash income in the saturated subcase collapses to  $\bar{\pi} + 1/(2\psi)$ , which is independent of productivity.

**Regime N: non-bunching above the threshold.** Firms with sufficiently high unconstrained profit do not bunch: the tax savings from locating at the threshold are too small relative to the real and reporting costs. They report truthfully, produce at the unconstrained level, and pay the statutory tax rate.

**The bunching boundary.** The upper boundary of the bunching region,  $\pi_{\max}^*(\psi)$ , is defined by the value of unconstrained profit at which bunching and paying tax yield the same payoff. When the reporting wedge saturates before the firm exits the bunching region (i.e.,  $\psi \geq \psi^{\min} \equiv (1 - 2\tau)/(2\tau\bar{\pi})$ ), the boundary takes the closed form

$$\pi_{\max}^*(\psi) = \frac{\bar{\pi} + 1/(2\psi)}{1 - \tau}. \quad (13)$$

Firms with low misreporting costs,  $\psi_i < \psi^{\min}$ , can use the reporting margin enough to avoid real output distortion throughout the bunching region. Firms with high misreporting costs,  $\psi_i \geq \psi^{\min}$ , exhaust the reporting margin sooner and therefore use both margins. Heterogeneity in  $\psi_i$  generates a non-degenerate composition of bunchers at the threshold.

## 5.5 Input-Share Implications

The model generates distinct predictions for reported labor and intermediate-input shares. Because misreporting flows only through reported labor, bunching mechanically raises the reported labor share. Intermediate inputs are affected only when firms reduce real output. A discontinuous *increase* in reported labor shares at the threshold is therefore evidence of reporting responses, while

a discontinuous *decline* in intermediate-input shares is evidence of real output distortion.

For Cobb–Douglas with  $\theta < 1$ , true labor and intermediate-input shares at output  $y$  are

$$\frac{\ell(y)}{y} = \beta \left( \frac{y}{y^*} \right)^{(1-\theta)/\theta}, \quad \frac{m(y)}{y} = \gamma \left( \frac{y}{y^*} \right)^{(1-\theta)/\theta}. \quad (14)$$

Define the real-shrinkage factor for saturated bunchers as  $\rho_i \equiv (\hat{y}(A_i, \psi_i)/y^*(A_i))^{(1-\theta)/\theta} \in (0, 1)$ .

Let  $\omega_{\text{sat}}$  denote the share of saturated bunchers among all bunchers and  $\bar{\rho} \equiv \mathbb{E}[\rho_i \mid \text{saturated bunching}]$ .

The discontinuities in input shares at the threshold can be written as

$$d_m \equiv \mathbb{E}[m/y \mid \text{bunch}] - \gamma = -\omega_{\text{sat}}\gamma(1 - \bar{\rho}), \quad (15)$$

$$d_\ell \equiv \mathbb{E}[\ell^r/y \mid \text{bunch}] - \beta = \frac{\beta}{\gamma}d_m + \mathbb{E}[\Delta/y \mid \text{bunch}]. \quad (16)$$

The labor-share discontinuity decomposes into a real-shrinkage component (proportional to  $d_m$ ) and an average reporting wedge  $\overline{\Delta/y}_{\text{bunch}} \equiv \mathbb{E}[\Delta/y \mid \text{bunch}]$ . The intermediate-input moment identifies the product  $\omega_{\text{sat}}(1 - \bar{\rho})$ , and the residual labor-share moment identifies  $\overline{\Delta/y}_{\text{bunch}}$ .

## 5.6 Dynamic Implications: Revenue Growth and Exit

The static model has dynamic implications once productivity evolves over time and firms face exit risk.

**Productivity evolution and revenue growth.** Suppose effective productivity follows  $\log A_{t+1} = \log A_t + \xi_t$ , with  $\xi_t \sim F$ . Each period, firms re-solve the static problem. Define revenue growth  $g_t \equiv \log y_{t+1} - \log y_t$ . For firms outside the saturated bunching region, output equals  $y^*(A) \propto A^{1/(1-\theta)}$ , so revenue growth has elasticity  $1/(1-\theta)$  with respect to productivity. For saturated bunchers, the constraint  $\pi(\hat{y}; A) = \bar{\pi} + 1/\psi_i$  pins true profit at a productivity-invariant level. Productivity gains must be offset by real output reductions to remain at the notch. The local elasticity of revenue with respect to productivity is therefore nonpositive,

$$\eta_{B\text{-sat},i} = -\frac{s_i}{1 - s_i} \leq 0, \quad s_i \equiv \frac{C(\hat{y}; A_i)}{\theta \hat{y}}. \quad (17)$$

Interior bunchers and non-bunchers grow at the unconstrained rate; only saturated bunchers exhibit attenuated growth. Letting  $\mu_\xi = \mathbb{E}[\xi_t]$ , the model implies a population-level growth gap

$$\mathbb{E}[g \mid \text{non-bunch}] - \mathbb{E}[g \mid \text{bunch}] = \omega_{\text{sat}} \left[ \frac{1}{1 - \theta} - \mathbb{E}[\eta_{B\text{-sat},i} \mid \text{saturated}] \right] \mu_\xi. \quad (18)$$

The growth gap is increasing in the share of saturated bunchers and in the extent to which the notch attenuates their growth.

**Exit.** After production, the firm draws an outside-option shock  $\zeta_t \sim \text{Exp}(1/\bar{\zeta})$  and exits if  $\zeta_t$  exceeds net cash income  $\Pi_t$ . The exit hazard is  $h(\Pi_t) = \exp(-\Pi_t/\bar{\zeta})$ , decreasing in current net cash. For non-bunchers above the threshold, net cash is  $(1 - \tau)\pi^*$ , so exit declines with productivity. For saturated bunchers, net cash is the productivity-invariant  $\bar{\pi} + 1/(2\psi_i)$ , so their exit hazard is independent of productivity. The same constraint that attenuates revenue growth among saturated bunchers also weakens the productivity-survival link. Assuming  $\pi^* \sim \text{Exp}(\lambda)$  independently of  $\psi$ , the population hazard ratio between saturated bunchers and non-bunchers is

$$\frac{\bar{h}_{\text{sat}}}{\bar{h}_N} = 1 + \frac{1 - \tau}{\lambda \bar{\zeta}}. \quad (19)$$

The proportional excess exit hazard identifies the composite outside-option scale  $\lambda \bar{\zeta}$ .

## 5.7 Joint Identification

The model links input shares, revenue growth, and exit through a small set of population-average parameters: the share of saturated bunchers  $\omega_{\text{sat}}$ , the mean real-shrinkage factor  $\bar{\rho}$ , the average reporting wedge  $\overline{\Delta/y}_{\text{bunch}}$ , and the outside-option scale  $\lambda \bar{\zeta}$ . The intermediate-input moment  $d_m$  identifies  $\omega_{\text{sat}}(1 - \bar{\rho})$ . The labor-share moment  $d_\ell$ , combined with  $d_m$ , identifies the average reporting wedge  $\overline{\Delta/y}_{\text{bunch}}$ . The revenue-growth gap identifies  $\omega_{\text{sat}}$ , and combined with  $d_m$  separates  $\bar{\rho}$  from  $\omega_{\text{sat}}$ . The exit-hazard gap identifies  $\lambda \bar{\zeta}$ .

The empirical patterns documented in Section 4—a small drop in  $m/y$  and a larger increase in  $\ell^r/y$  at the threshold, lower revenue growth and higher exit among bunchers—therefore map directly to the model’s bunching, real-output, growth, and exit channels.

## 5.8 Limitations

The simple model intentionally omits several features that the structural extension developed in the appendix will incorporate. First, all bunchers locate exactly at the threshold, whereas the data exhibit diffuse bunching with some firms just above  $\bar{\pi}$ . Introducing an idiosyncratic fixed cost of misreporting smooths the bunching mass. Second, the labor margin treats family and non-family workers symmetrically, even though Section 4 shows that profit shifting flows primarily through family members. The model accommodates this by allowing the misreporting curvature to differ across labor types. Third, capital is fixed within the period; the dynamic structural extension introduces capital accumulation, deposits, and adjustment costs. These extensions are necessary for quantitative welfare analysis of policy counterfactuals, but the simple model already captures the core mechanisms.

## 6 Calibration

We calibrate the model to four empirical moments documented in Section 4. Together with auxiliary parameters from the production technology and sample averages, these moments pin down the population-average objects described in Section 5.7.

### 6.1 Empirical Moments

The target moments are: the intermediate-input share discontinuity at the threshold  $d_m \approx -2$  pp, the labor-share discontinuity  $d_\ell \approx +7$  pp, the revenue-growth gap of 2.1 pp ( $\approx 20\%$  of the sample mean of 11%), and the proportional excess exit hazard of 8.1% for bunchers. From the empirical sample we further use the labor cost share  $\beta = 0.16$  and the intermediate-input cost share  $\gamma = 0.75$ , implying the variable-input share  $\theta = 0.91$ . We set the post-reform statutory tax rate  $\tau = 0.17$  and the threshold  $\bar{\pi} = 12$  in units of 10,000 NTD, matching the 2010 reform.

### 6.2 Calibration via the Identification Equations

**Average reporting wedge.** The input-share moments separate the reporting response from the real-output response. From Equation (16), the average reporting wedge at the bunching mass is

$$\overline{\Delta/y}_{\text{bunch}} = d_\ell - \frac{\beta}{\gamma} d_m \approx 0.07 - \frac{0.16}{0.75} \cdot (-0.02) \approx 0.074.$$

Bunchers inflate reported labor expenses by approximately 7.4% of revenue on average. The intermediate-input moment further identifies  $\omega_{\text{sat}}(1 - \bar{\rho}) \approx 0.027$ .

**Saturated-buncher share.** The growth-gap equation (18) identifies the share of saturated bunchers. Using the empirical mean revenue growth of 11% as the non-bunching benchmark and the limiting case in which saturated bunchers have near-zero local growth elasticity, the observed gap of 2.1 pp implies  $\omega_{\text{sat}} \approx 0.19$ .

**Real-shrinkage factor.** Combining  $\omega_{\text{sat}} \approx 0.19$  with  $\omega_{\text{sat}}(1 - \bar{\rho}) \approx 0.027$  yields  $\bar{\rho} = 1 - 0.027/0.19 \approx 0.86$ . In output terms,  $\hat{y}/y^* = \bar{\rho}^{\theta/(1-\theta)} \approx 0.22$ . The implied output reduction is substantial for the subset of firms operating at the mixed margin.

**Outside-option scale.** Equation (19) maps the proportional excess hazard of 8.1% to  $\lambda\bar{\zeta} = (1 - \tau)/0.081 \approx 10.3$ . We anchor  $\bar{\zeta}$  separately using the sample-mean exit rate of 10.7% as the population average  $\bar{h}_{\text{sat}}$ .

### 6.3 Calibration Summary

Table 2 summarizes the calibrated population-average parameters and their identifying moments.

**Table 2:** Calibrated Parameters of the Simple Model

| Parameter                  | Symbol                          | Value | Source moment                       |
|----------------------------|---------------------------------|-------|-------------------------------------|
| Variable-input share       | $\theta = \beta + \gamma$       | 0.91  | Cost shares                         |
| Statutory tax rate         | $\tau$                          | 0.17  | Statute, post-reform                |
| Tax threshold              | $\bar{\pi}$                     | 12    | Statute, post-reform                |
| Mixed-margin share         | $\omega_{\text{sat}}$           | 0.19  | Revenue growth gap                  |
| Mean real-shrinkage factor | $\bar{\rho}$                    | 0.86  | $d_m$ and growth gap                |
| Average reporting wedge    | $\bar{\Delta}/y_{\text{bunch}}$ | 0.074 | Labor and intermediate-input shares |
| Outside-option composite   | $\lambda\bar{\zeta}$            | 10.3  | Exit hazard gap                     |
| Average bunching hazard    | $\bar{h}_{\text{sat}}$          | 0.107 | Mean exit rate                      |

*Notes:* This table reports the calibrated population-average parameters of the simple model from Section 5 and their identifying empirical moments. All monetary quantities are expressed in units of 10,000 NTD.

The distribution of misreporting curvatures  $H$  is not separately identified from these four moments. A natural next step, pursued in the dynamic structural extension, is to impose a parametric

form for  $H$  and use additional moments such as the overall bunching mass, the dispersion of growth among bunchers, or the shape of the bunching density to discipline the full distribution.

## 7 Policy Simulations

We use the calibrated model to evaluate the welfare consequences of the 2010 reform and to characterize its distributional incidence. We simulate  $N = 50,000$  firms with productivity drawn from an exponential distribution with mean 15 (in units of 10,000 NTD) and misreporting curvature drawn from a log-normal distribution with  $\mathbb{E}[\log \psi] = \log 0.3$  and  $\sigma_{\log \psi} = 0.6$ . Under this specification, the simulated saturation share is  $\omega_{\text{sat}} \approx 0.20$ , close to the calibrated value.

### 7.1 Welfare Accounting

We decompose social welfare into a static (per-period) component and a dynamic (lifetime NPV) component. The static accounting separates tax revenue (a transfer from firms to government) from two deadweight-loss components: the misreporting cost  $g(\Delta; \psi) = (\psi/2)\Delta^2$ , a real resource cost; and the real-output cost  $\pi^*(A) - \pi(\hat{y}; A)$ , the forgone profit at saturated bunchers operating below  $y^*$ .

To capture the dynamic channels documented in Section 4, we compute a 20-year discounted social value per firm. Each firm draws an exit shock in every period of survival, with hazard  $h_i = \exp(-\Pi_i/\bar{\zeta})$ . Conditional on surviving, per-period flow grows at the unconstrained rate  $g_{\text{unc}} = 0.03$  for non-saturated firms and stays constant for saturated bunchers, whose flow is pinned by the notch. Lifetime social value is

$$V_i^{\text{soc}} = (\Pi_i + \text{tax}_i) \sum_{t=0}^{T-1} [\delta(1 - h_i)(1 + g_i)]^t, \quad (20)$$

with discount factor  $\delta = 0.95$  and horizon  $T = 20$ . The lifetime deadweight loss is  $V_i^{\text{FB}} - V_i^{\text{soc}}$ , where the first-best value uses  $\pi_i^*$  as the flow and  $h_i^{\text{FB}} = \exp(-\pi_i^*/\bar{\zeta})$  as the hazard.

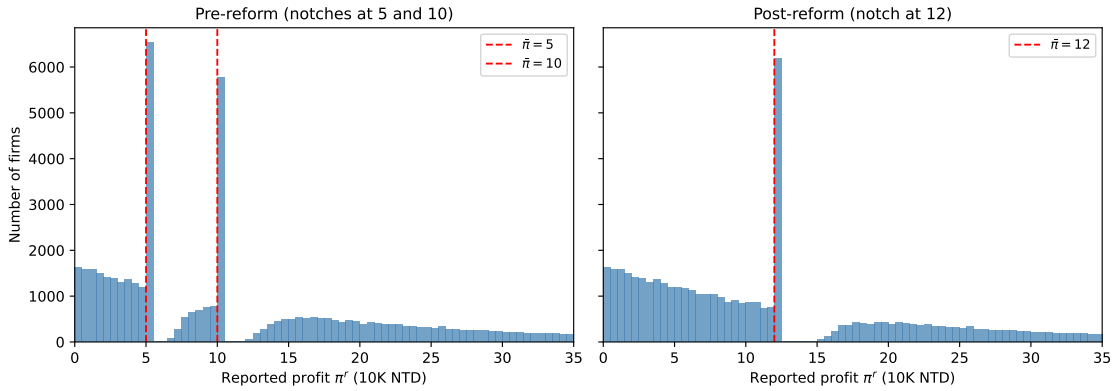
### 7.2 The 2010 Reform: Pre vs. Post Comparison

We hold the firm distribution fixed and replace the post-reform schedule with the pre-reform schedule: a notch at  $\bar{\pi}_1 = 5$  where the average tax rate jumps from 0 to 15%, followed by another notch

at  $\bar{\pi}_2 = 10$  where the rate jumps to 25%. Under the pre-reform schedule, firms choose among (i) operating unconstrained and paying tax according to their bracket, (ii) bunching at the first notch, or (iii) bunching at the second notch.

**Distribution of reported profit.** Figure 6 compares the simulated distribution of reported profit before and after the reform. Pre-reform, a large bunching mass appears at  $\pi^r = 5$  and a much smaller mass at  $\pi^r = 10$ . Post-reform, the bunching mass concentrates at the new threshold of 12. The simulated patterns mirror the empirical evidence in Figure 2: the bunching mass shifts immediately and entirely to the new exemption threshold.

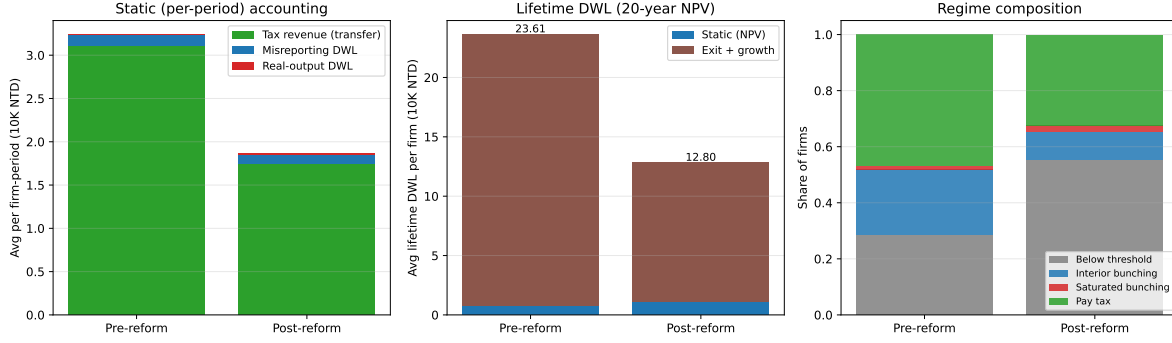
**Figure 6: Simulated Distribution of Reported Profit: Pre vs. Post Reform**



*Notes:* This figure shows the simulated distribution of reported profit under the pre-reform (left) and post-reform (right) schedules. The pre-reform schedule has notches at  $\bar{\pi}_1 = 5$  (15%) and  $\bar{\pi}_2 = 10$  (25%). The post-reform schedule has a single notch at  $\bar{\pi} = 12$  (17%). All monetary quantities are expressed in units of 10,000 NTD.

**Welfare decomposition.** Figure 7 compares the static and lifetime welfare consequences of the reform. Table 3 reports the underlying per-firm averages.

Three findings stand out. First, the reform reduces tax revenue by roughly 44% per firm because the new threshold at  $\bar{\pi} = 12$  exempts most firms with unconstrained profit below 12, while the pre-reform schedule taxed firms above  $\bar{\pi}_1 = 5$ . Second, the reform reduces static misreporting DWL but increases the static real-output DWL. The misreporting reduction reflects the simpler schedule. The real-output increase reflects the location of the new threshold: with  $\bar{\pi} = 12$  and typical  $1/\psi \approx 3$ , only firms with relatively high unconstrained profit reach the saturated bunching region. Combined with the modestly higher rate of 17% relative to the binding pre-reform rate of 15% at

**Figure 7: Pre vs. Post Reform Welfare Comparison**

*Notes:* The left panel decomposes per-firm static accounting into tax revenue (a transfer to the government) and the two deadweight-loss components (misreporting and real output). The middle panel decomposes the 20-year lifetime DWL into a static-NPV component and the additional exit-plus-growth amplification. The right panel reports the composition of firms by regime.

**Table 3: Per-Firm Averages: Pre-Reform vs. Post-Reform**

| Metric (per firm)                 | Pre-reform | Post-reform | Change |
|-----------------------------------|------------|-------------|--------|
| Tax revenue                       | 3.11       | 1.75        | −1.36  |
| <i>Static (per-period) DWL</i>    |            |             |        |
| Misreporting DWL                  | 0.126      | 0.099       | −0.027 |
| Real-output DWL                   | 0.003      | 0.018       | +0.015 |
| Total static DWL                  | 0.129      | 0.117       | −0.012 |
| <i>Lifetime (20-year NPV) DWL</i> |            |             |        |
| Static NPV part                   | 0.76       | 1.09        | +0.32  |
| Exit-plus-growth part             | 22.85      | 11.71       | −11.14 |
| Total lifetime DWL                | 23.61      | 12.80       | −10.82 |
| Average exit hazard               | 0.313      | 0.293       | −0.020 |
| Average lifetime social value $V$ | 155.6      | 166.4       | +10.8  |

*Notes:* This table reports per-firm averages of welfare components under the pre-reform and post-reform schedules. All monetary quantities are in units of 10,000 NTD.

the lower notch, more firms move into saturated bunching post-reform. Third, the reform substantially reduces lifetime DWL, with the bulk of the saving coming from the exit-plus-growth channel. The mechanism is that the higher threshold moves many firms from the tax-paying regime—where elevated exit hazard compounds over the 20-year horizon—into the below-threshold regime, where  $\Pi = \pi^*$  and the exit hazard equals its first-best value. The dynamic margin therefore dominates the welfare improvement of the reform, consistent with the empirical pattern of attenuated growth

and elevated exit among bunchers.

### 7.3 Distributional Incidence

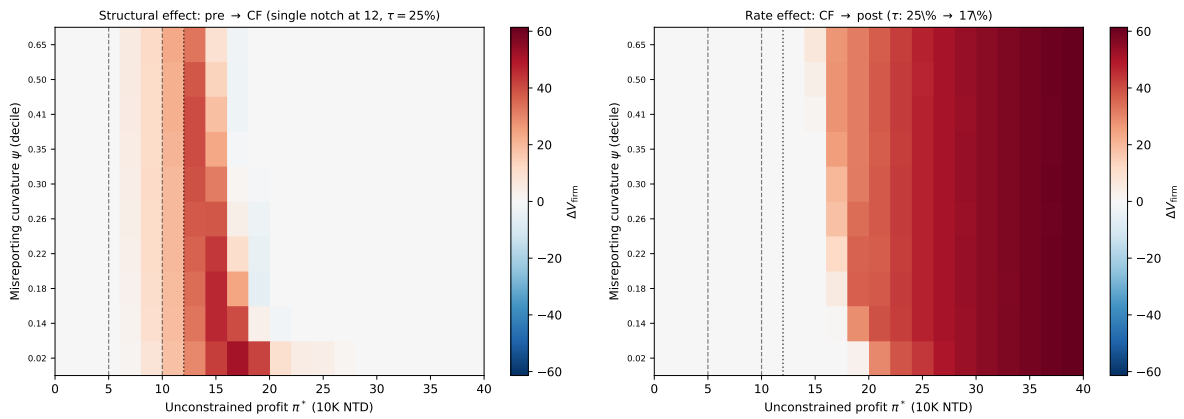
We examine which firms benefit from the reform, splitting by unconstrained profit  $\pi^*$  and misreporting curvature  $\psi$ . A first observation is that the reform is Pareto-improving on the firm side. The post-reform tax schedule weakly dominates the pre-reform schedule at every reported-profit value:  $T^{\text{post}}(\pi^r) \leq T^{\text{pre}}(\pi^r)$  for all  $\pi^r$ . For any feasible  $(y, \Delta)$  pair, the firm's post-reform payoff weakly exceeds its pre-reform payoff. The simulation confirms that no firm is made worse off: 71.6% of firms strictly gain in lifetime firm value, 28.4% are unaffected (those with  $\pi^* < 5$ ), and 0% lose.

To separate the structural change (threshold and schedule simplification) from the mechanical rate change (25% to 17%), we construct a counterfactual schedule with the post-reform structure (single notch at  $\bar{\pi} = 12$ ) and the pre-reform top rate (25%). The total reform impact decomposes as

$$\underbrace{\Delta V_{\text{firm}}^{\text{total}}}_{\text{pre} \rightarrow \text{post}} = \underbrace{\Delta V_{\text{firm}}^{\text{structural}}}_{\text{pre} \rightarrow \text{CF}} + \underbrace{\Delta V_{\text{firm}}^{\text{rate}}}_{\text{CF} \rightarrow \text{post}}.$$

On average, the structural effect accounts for 6.76 units of lifetime firm value (29% of the total), and the rate effect accounts for 16.62 units (71%). The bulk of the headline welfare gain reflects the mechanical rate cut. The structural component is where the model's behavioral predictions matter.

**Figure 8:** Distributional Decomposition: Structural vs. Rate Effects



*Notes:* This figure decomposes the lifetime firm-value gain from the 2010 reform into a structural effect (left panel: pre-reform → counterfactual single notch at  $\bar{\pi} = 12$  with the pre-reform 25% rate) and a rate effect (right panel: counterfactual → post-reform,  $\tau : 25\% \rightarrow 17\%$ ). Vertical lines mark the three notches. Color intensity represents the per-firm gain in lifetime value.

Figure 8 shows the structural and rate effects separately across the  $(\pi^*, \psi)$  plane. The two heatmaps look qualitatively different. The structural effect is concentrated among firms with unconstrained profit in the range  $\pi^* \in [5, 12]$ . These firms were tax-paying or bunching under the pre-reform schedule because their profit exceeded the lower notch at 5, but they sit below the new threshold of 12. They move from paying tax (or paying misreporting cost) to operating unconstrained at no cost. The  $\psi$  gradient is visible: among firms in this productivity range, low- $\psi$  firms (cheap misreporting) had bunched less efficiently under the old schedule and gain more when the bunching opportunity disappears. Firms with  $\pi^*$  well above 12 see almost no structural gain because they still pay tax under the counterfactual.

The rate effect is concentrated among firms with high unconstrained profit,  $\pi^* > 12$ . These firms pay tax in both regimes, and the lower statutory rate mechanically increases their net cash income by approximately  $(0.25 - 0.17) \cdot \pi^* = 0.08 \pi^*$  per period. The effect grows linearly with productivity. The  $\psi$  dimension contributes essentially nothing because the rate change has no behavioral component for tax-paying firms.

Stripping out the mechanical rate cut, the model therefore predicts that the structural change primarily benefits medium-productivity firms—those whose unconstrained profit lies between the old and new thresholds. Within this group, the  $\psi$  dimension contributes a modest secondary gradient. Productivity is the first-order determinant of distributional incidence; misreporting curvature is a second-order modifier.

## 7.4 Discussion

Two implications are worth highlighting. First, the simple model rationalizes the empirical patterns of Section 4 through a tightly linked set of mechanisms. The discontinuities in input shares identify the average reporting wedge and the product of the saturated-buncher share and real-output distortion. The revenue growth gap identifies the saturated-buncher share, and combined with the input-share moments, separately identifies the real-shrinkage factor. The exit-hazard gap identifies the outside-option scale. The same parameters that explain bunching also explain the lower revenue growth and higher exit among bunching firms.

Second, the simulation suggests that the 2010 reform’s main welfare gain operates through the dynamic exit margin. Approximately 97% of the lifetime welfare improvement comes from the exit-plus-growth channel: the higher threshold moves firms from the tax-paying region (where elevated exit risk compounds over time) into the unconstrained region. This finding emphasizes that

static analyses understate the welfare cost of notched schedules when firm dynamics are sensitive to net cash income. The structural decomposition further shows that medium-productivity firms with unconstrained profit between the old and new thresholds are the principal beneficiaries of the threshold change, while high-productivity firms benefit primarily from the mechanical rate cut. The reform thus combines a behaviorally meaningful structural component with a larger but more mechanical rate component, and most of the firm-side welfare gain accrues to medium-productivity firms.

## 8 Conclusion

In this paper, we examine firms' responses and the aggregate policy implications of notched corporate income tax design, using evidence from a corporate income tax reform in Taiwan. Empirically, we find that firms respond strongly to changes in the exemption threshold on the corporate income tax schedule, employing a mix of real and reporting strategies. On one hand, firms over-report salaries for employees who are family members of the firm owner, inflating deductions to lower taxable income. On the other hand, firms strategically reduce production and slow down their revenue growth. The stagnant growth further leads to an unexpected increase in exit probability.

To rationalize these patterns and quantify the welfare consequences of the reform, we develop a parsimonious static model that links the static input-share distortions to the dynamic patterns of lower revenue growth and higher exit through a small set of population-average parameters. The model is identified by four empirical moments documented in our reduced-form analysis: the discontinuities in the intermediate-input and labor shares at the threshold, the revenue growth gap, and the excess exit hazard. Heterogeneity in firms' misreporting costs is essential: low-curvature firms bunch primarily through reporting, while high-curvature firms exhaust the reporting margin and reduce real output. The latter group is responsible for the dynamic patterns: their flow is pinned at a productivity-invariant level, attenuating their growth and raising their exit hazard.

Using the calibrated model, we simulate the 2010 reform and decompose its welfare consequences. The reform substantially reduces lifetime deadweight loss, with most of the gain operating through the dynamic exit margin rather than through static reductions in bunching distortion. Decomposing the reform into a structural change (raising the threshold and simplifying the schedule) and a mechanical rate change (25% to 17%), we find that the structural change disproportionately benefits medium-productivity firms whose unconstrained profit lies between the old and new

thresholds, while the rate change is a more uniform transfer to firms paying tax in both regimes. No firms are made worse off by the reform: the post-reform schedule weakly dominates the pre-reform schedule at every reported-profit value.

## References

- Miguel Almunia and David Lopez-Rodriguez. Under the radar: The effects of monitoring firms on tax compliance. *American Economic Journal: Economic Policy*, 10(1):1–38, 2018.
- Amrit Amirapu and Michael Gechter. Labor Regulations and the Cost of Corruption: Evidence from the Indian Firm Size Distribution. *The Review of Economics and Statistics*, 102(1):34–48, March 2020.
- Santosh Anagol, Allan Davids, Benjamin B Lockwood, and Tarun Ramadorai. Diffuse bunching with frictions: Theory and estimation. *NBER Working Paper*, (w32597), 2024.
- Alan J Auerbach, James R Hines Jr, and Joel Slemrod. *Taxing corporate income in the 21st century*. Cambridge University Press, 2007.
- Bee Yan Aw, Sukkyun Chung, and Mark J Roberts. Productivity, output, and failure: a comparison of taiwanese and korean manufacturers. *The Economic Journal*, 113(491):F485–F510, 2003.
- Pierre Bachas and Mauricio Soto. Corporate taxation under weak enforcement. *American Economic Journal: Economic Policy*, 13(4):36–71, 2021.
- Marinho Bertanha, Andrew H McCallum, and Nathan Seegert. Better bunching, nicer notching. *Journal of Econometrics*, 237(2):105512, 2023.
- Serguey Braguinsky, Lee G Branstetter, and Andre Regateiro. The incredible shrinking portuguese firm. Technical report, National Bureau of Economic Research, 2011.
- Paul Carrillo, Dina Pomeranz, and Monica Singhal. Dodging the taxman: Firm misreporting and limits to tax enforcement. *American Economic Journal: Applied Economics*, 9(2):144–164, 2017.
- Zhao Chen, Zhikuo Liu, Juan Carlos Suárez Serrato, and Daniel Yi Xu. Notching r&d investment with corporate income tax cuts in china. *American Economic Review*, 111(7):2065–2100, 2021.
- Raj Chetty, John N Friedman, Tore Olsen, and Luigi Pistaferri. Adjustment costs, firm responses, and micro vs. macro labor supply elasticities: Evidence from danish tax records. *The quarterly journal of economics*, 126(2):749–804, 2011.
- Keshav Choudhary and Bhanu Gupta. Dynamics of firm growth around tax thresholds: Evidence from india. 2024.
- Jeffrey L Coles, Elena Patel, Nathan Seegert, and Matthew Smith. How do firms respond to corporate taxes? *Journal of Accounting Research*, 60(3):965–1006, 2022.

- Michael P Devereux, Li Liu, and Simon Loretz. The elasticity of corporate taxable income: New evidence from uk tax records. *American Economic Journal: Economic Policy*, 6(2):19–53, 2014.
- Dhammika Dharmapala, Joel Slemrod, and John Douglas Wilson. Tax policy and the missing middle: Optimal tax remittance with firm-level administrative costs. *Journal of Public Economics*, 95(9-10):1036–1047, 2011.
- Rebecca Diamond and Petra Persson. The long-term consequences of teacher discretion in grading of high-stakes tests. Technical report, National Bureau of Economic Research, 2016.
- Nadja Dwenger and Viktor Steiner. Profit taxation and the elasticity of the corporate income tax base: Evidence from german corporate tax return data. *National Tax Journal*, 65(1):117–150, 2012.
- Bertrand Garbinti, Jonathan Goupille-Lebret, Mathilde Muñoz, Stefanie Stantcheva, and Gabriel Zucman. Tax design, information, and elasticities: evidence from the french wealth tax. Technical report, National Bureau of Economic Research, 2023.
- Luis Garicano, Claire Lelarge, and John Van Reenen. Firm size distortions and the productivity distribution: Evidence from france. *American Economic Review*, 106(11):3439–79, 2016.
- François Gourio and Nicolas Roys. Size-dependent regulations, firm size distribution, and reallocation. *Quantitative Economics*, 5(2):377–416, 2014.
- Patrick Kennedy, Christine Dobridge, Paul Landefeld, and Jacob Mortenson. The efficiency-equity tradeoff of the corporate income tax: Evidence from the tax cuts and jobs act. *Unpublished manuscript*, 2022.
- Henrik J Kleven and Mazhar Waseem. Using notches to uncover optimization frictions and structural elasticities: Theory and evidence from pakistan. *The Quarterly Journal of Economics*, 128(2):669–723, 2013.
- Henrik Jacobsen Kleven, Martin B Knudsen, Claus Thustrup Kreiner, Søren Pedersen, and Emmanuel Saez. Unwilling or unable to cheat? evidence from a tax audit experiment in denmark. *Econometrica*, 79(3):651–692, 2011.
- Li Liu, Benjamin Lockwood, and Eddy Tam. Small firm growth and the vat threshold: evidence for the uk. 2022.
- Felipe Lobel, Thiago Scot, and Pedro Zúniga. Corporate taxation and evasion responses: Evidence from a minimum tax in honduras. *American Economic Journal: Economic Policy*, 16(1):

482–517, 2024.

Athiphat Muthitacharoen, Wonma Wanichthaworn, and Trongwut Burong. Vat threshold and small business behavior: evidence from thai tax returns. *International Tax and Public Finance*, 28(5): 1242–1275, 2021.

Emmanuel Saez. Do taxpayers bunch at kink points? *American economic Journal: economic policy*, 2(3):180–212, 2010.

Dustin Swonder and Damián Vergara. A sufficient statistics approach to optimal corporate taxes. 2025.

# Supplemental Appendix for Notch and Catch: Dynamic Inefficiency in the Corporate Income Tax Schedule

## Table of Contents

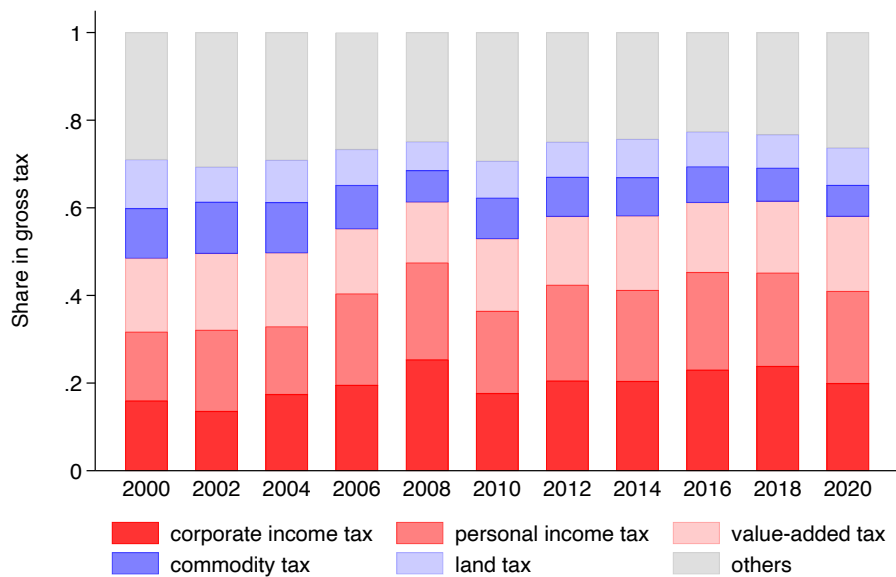
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|                                            |     |
|--------------------------------------------|-----|
| A Additional Figures. . . . .              | A1  |
| B Bunching Estimators. . . . .             | A9  |
| C Alternative Empirical Strategy . . . . . | A12 |

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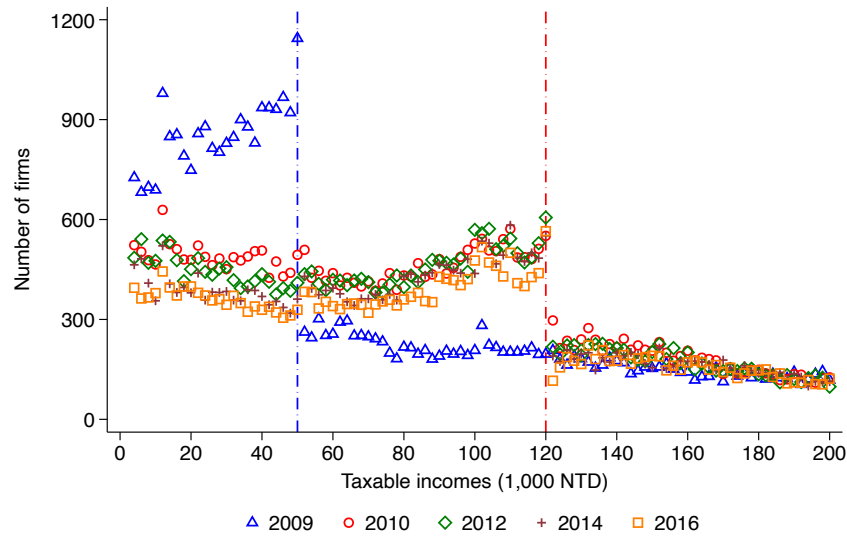
## A Additional Figures

**Figure A.1:** Different Sources of Gross Tax Revenue in Taiwan 2000-2020



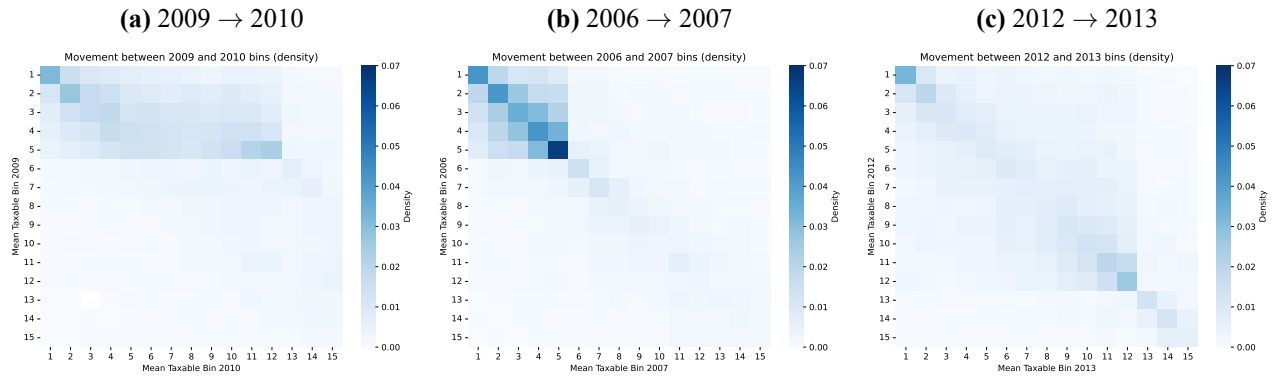
*Notes:* This figure plots the shares of different types of taxes to gross tax revenues in Taiwan between 2000 and 2020. The data source is the Public Finance Statistics Database in Taiwan accessed at <https://web02.mof.gov.tw/njswww/webmain.aspx?sys=100&funid=edefjspf2>.

**Figure A.2:** Distribution of Corporate Taxable Incomes by Years



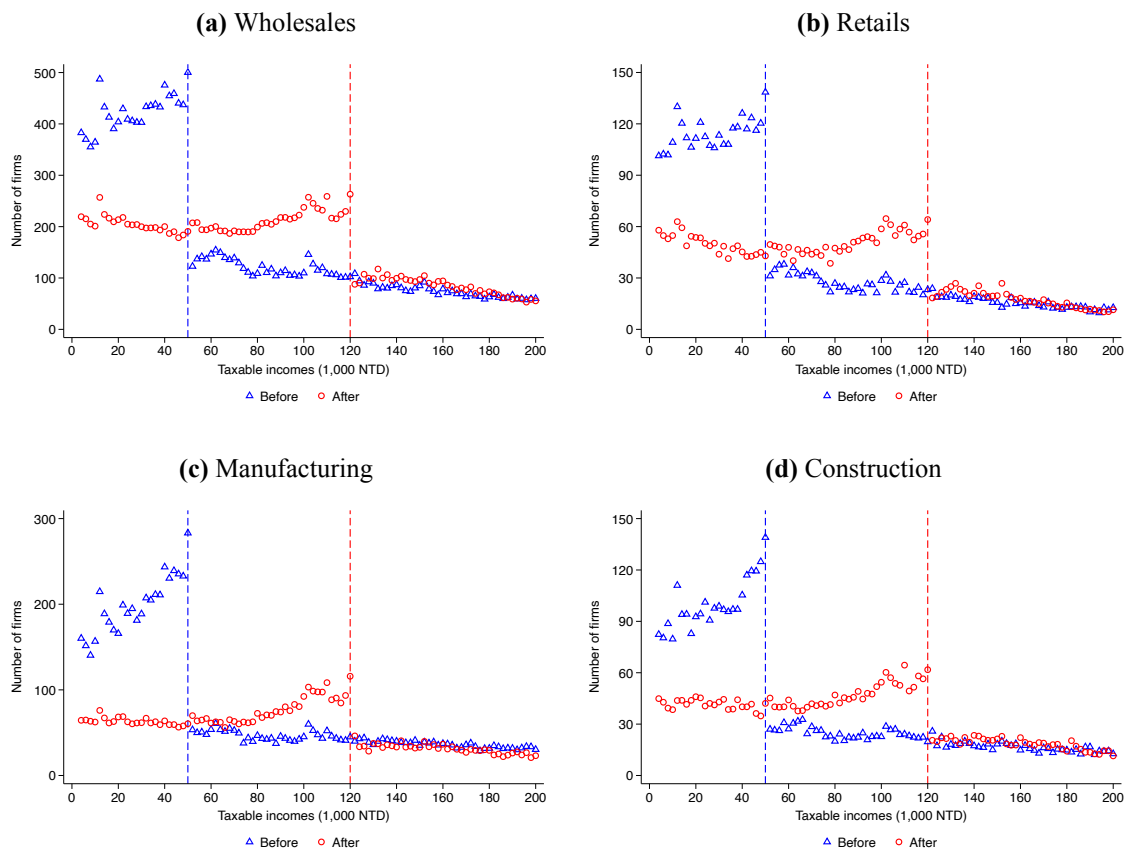
*Notes:* This figure shows the average annual distribution of corporate taxable incomes across years between 2009 and 2016. The vertical axis denotes the average number of firms yearly in each taxable income bin. The horizontal axis represents the taxable income bins.

**Figure A.3:** Density of Firms Moving across Taxable Income Bins



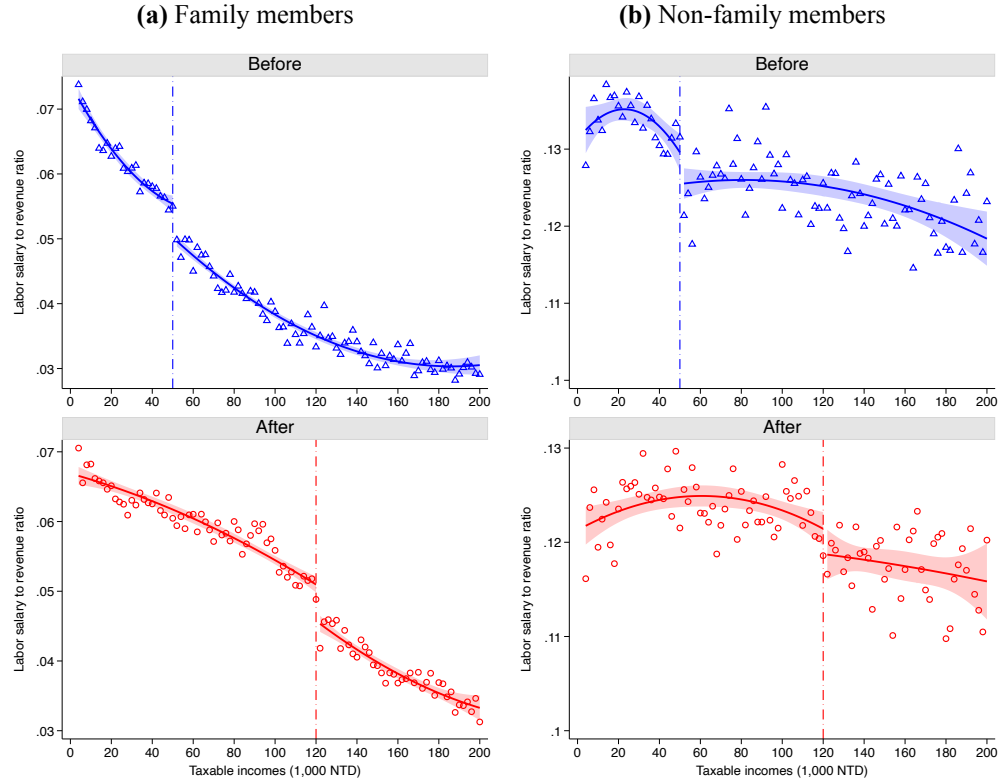
*Notes:* This figure presents a heatmap of firm transitions between taxable income bins in consecutive years. The horizontal axis denotes the average taxable income bin in year  $t$ , and the vertical axis shows the corresponding bin in year  $t + 1$ . Darker shading indicates a higher number of firms moving between the two bins. The diagonal reflects firms that remain in the same income bin across years, while off-diagonal cells capture upward or downward mobility. Panel (a) illustrates firm movements for 2009–2010, while Panels (b) and (c) plots transitions in 2006–2007 and 2012–2013 for comparison.

**Figure A.4: Distribution of Corporate Taxable Incomes by Years and by Sectors**



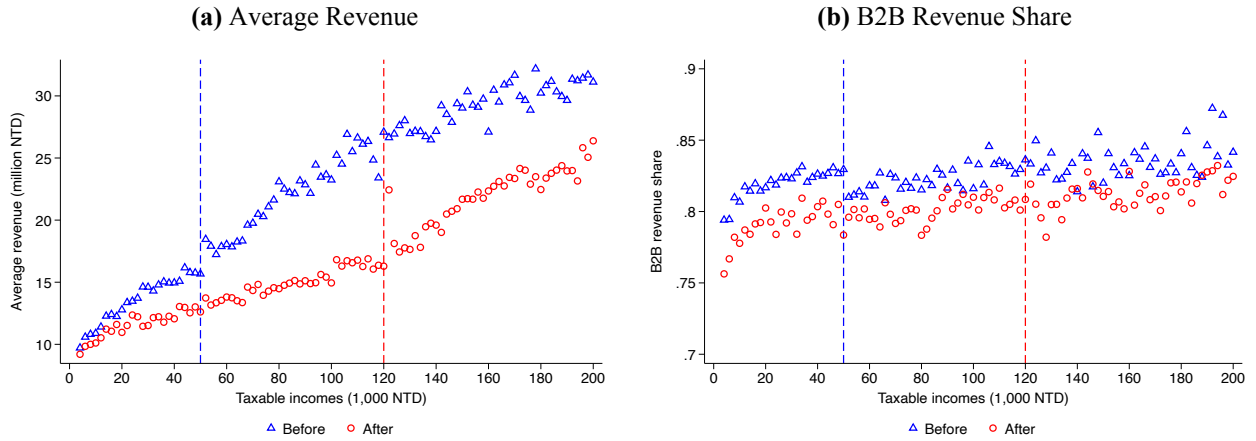
*Notes:* This figure shows the average annual distribution of corporate taxable incomes across sectors. Panels (a)-(d) plot the average annual distribution of corporate taxable incomes in wholesales, retails, manufacturing, and construction sectors before the reform (2004-2009, in blue triangles) and after (2010-2017, in red circles). The vertical axis denotes the average number of firms yearly in each taxable income bin. The horizontal axis represents the taxable income bins.

**Figure A.5: Labor Salary Ratio and Taxable Income: Family and Non-family Employees**



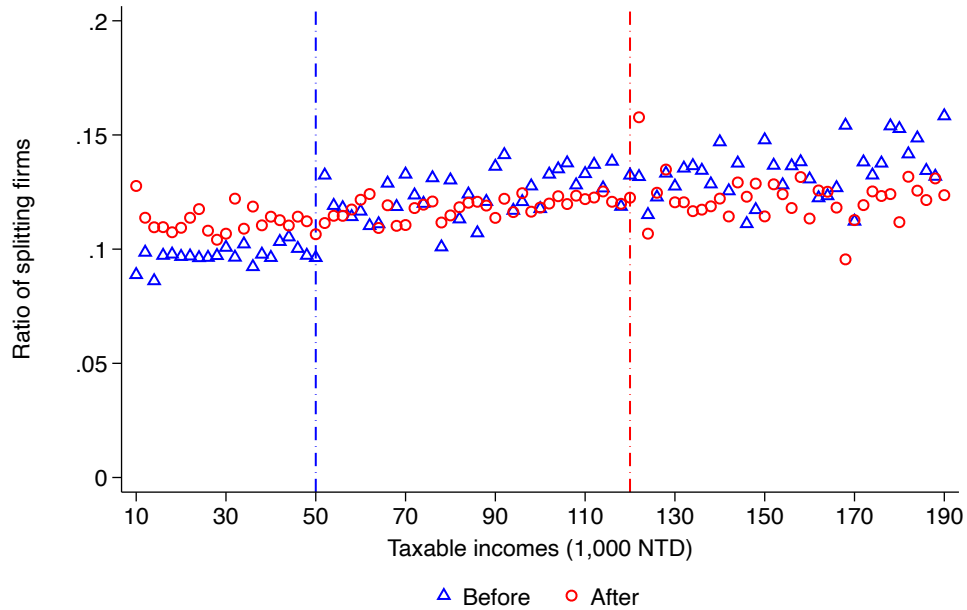
*Notes:* This figure shows the ratios of labor salary and gross revenue with respect to corporate taxable income bins before the reform (2004-2009, upper panels with blue triangles and lines) and after (2010-2017, lower panels with red circles and lines). Panel (a) plots the ratios of labor salary paid to family members of firm owners and gross revenue; Panel (b) plots the ratios of labor salary paid to non-family-member employees and gross revenue. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

**Figure A.6: How Firms Bunch: Alternative Channels**



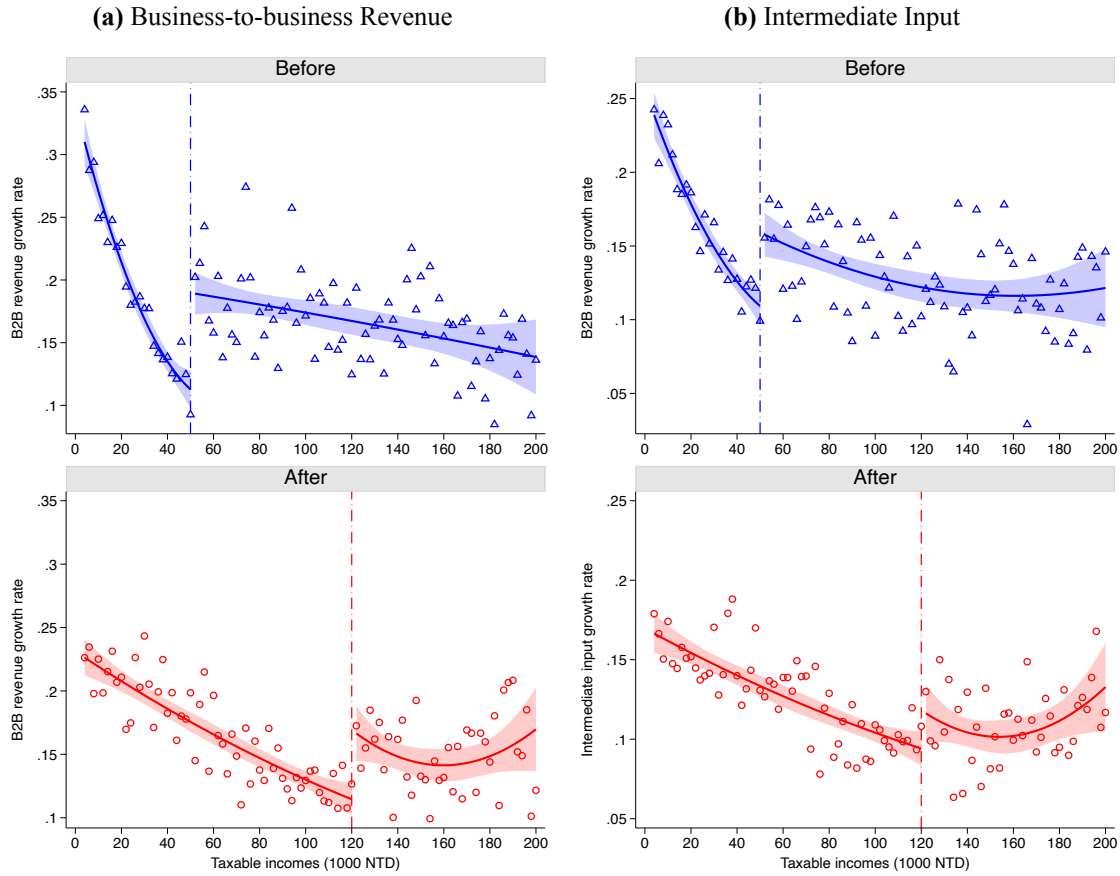
*Notes:* This figure examines alternative channels for firms to achieve bunching. Panel (a) plots the average firm revenues with respect to corporate taxable income bins before the reform (2004–2009, blue triangles) and after (2010–2017, red circles). Panel (b) plots the shares of business-to-business revenues in gross revenues.

**Figure A.7: Ratio of Firm Splitting by Taxable Incomes**



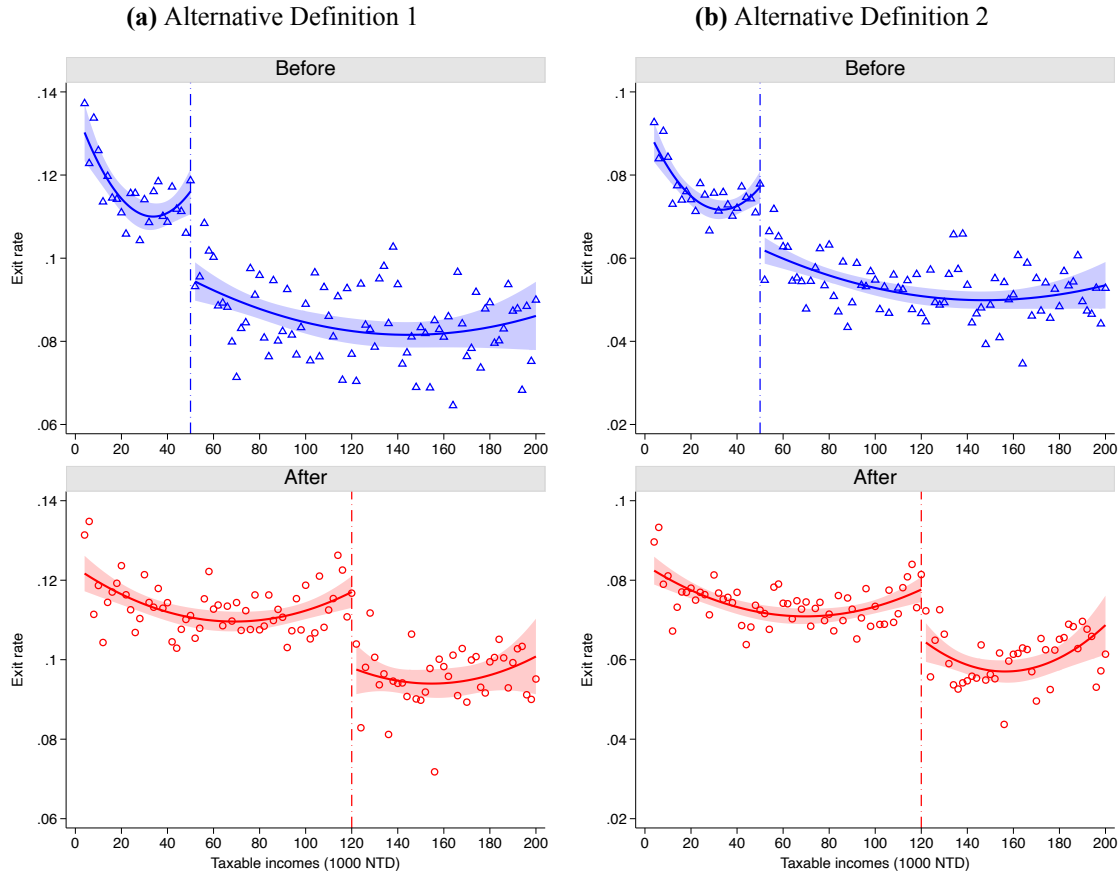
*Notes:* This figure shows the ratio of splitting firms in each taxable income bin before the reform (2004–2009, in blue circle) and after (2010–2017, in red plus). The vertical axis denotes the ratio of splitting firms to the total number of firms in each taxable income bin. Splitting firms are defined as those which have a common shareholder with any other firms. The horizontal axis represents the taxable income bins. For example, the 5<sup>th</sup> bin represents firms that have taxable income between 50 thousand NTD (included) and 60 thousand NTD (not included).

**Figure A.8: Business-to-business Revenue Growth Rate and Taxable Income**



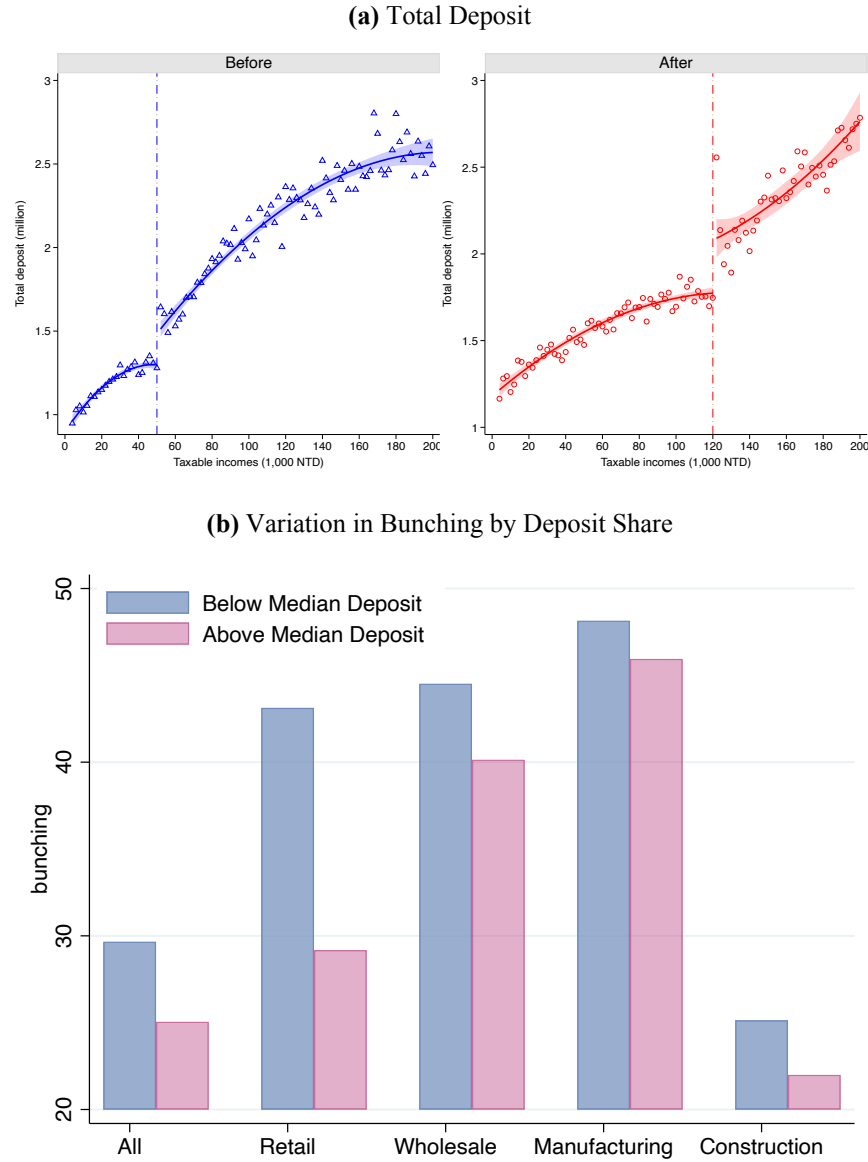
*Notes:* This figure shows the firm growth rate with respect to corporate taxable income bins before the reform (2004-2009, upper panels with blue triangles and lines) and after (2010-2017, lower panels with red circles and lines). Panel (a) plots the growth rate of firm revenues from business-to-business transactions; Panel (b) plots the growth rate of firm intermediate inputs. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

**Figure A.9: Exit Rate and Taxable Income: Robustness Checks**



*Notes:* This figure shows the average exit rates with respect to corporate taxable income bins before the reform (2004-2009, upper panels with blue triangles and lines) and after (2010-2017, lower panels with red circles and lines). Panel (a) plots the exit rate where firm exit is defined as missing reported tax records; Panel (b) plots the exit rate where firm exit is defined as absent from the tax records for three consecutive years. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

**Figure A.10: Liquidity Constraints and Taxable Income**



*Notes:* This figure shows how total deposits vary with taxable incomes. Panel (a) plots total deposits with respect to corporate taxable income bins before the reform (2004-2009) and after (2010-2017). The horizontal axis represents the taxable income bins. The vertical axis denotes the firms' total deposits. The vertical dashed line indicates the corporate income tax exemption threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

## B Bunching Estimators

We adopt the bunching estimator following [Saez \(2010\)](#) and [Kleven and Waseem \(2013\)](#) to quantify firms' bunching response to the tax notches after the reform.<sup>14</sup> We first group the data into bins of the firms' corporate taxable income  $x_j$ , and estimate the counterfactual distribution using the following equation.

$$f_j = \sum_{i=0}^q \beta_i \cdot (x_j)^i + \sum_{k=x_{ub}}^{x_{lb}} \gamma_k \cdot \mathbb{1}(x_j = k) + \eta_j. \quad (21)$$

$f_j$  is the number of firms in taxable income bin  $j$ . We use  $q$  to denote the order of the polynomials. By assuming that only firms within the exclusion region  $[x_{ub}, x_{lb}]$  would respond, we can use the distribution outside of the exclusion region to construct the counterfactual.  $\gamma_k$  captures the intercept shift generated by firms' response to the policy of each bin within the exclusion region. The counterfactual distribution, in the absence of the policy, is predicted by the follows.

$$\hat{f}_j = \sum_{i=0}^q \hat{\beta}_i \cdot (x_j)^i.$$

We denote the tax threshold as  $\bar{x}$  and the bunching mass area ( $X_B$ ) and the missing mass area ( $X_H$ ) are given by the follows.

$$\hat{X}_B = \sum_{j=x_{lb}}^{\bar{x}} (f_j - \hat{f}_j) \geq 0, \quad \hat{X}_H = \sum_{j=\bar{x}}^{x_{ub}} (\hat{f}_j - f_j) \geq 0.$$

The estimates of the bunching response are shown in Figure [B.1](#). The solid line represents the estimated counterfactual distribution. The vertical gray line indicates the lower and upper bound of the exclusion region, generated by a data-driven procedure to ensure that the aggregated density of observed and counterfactual distribution is the same. Firms' bunching response  $dI$  is defined as the percentage of the firms' density in the exclusive region, which measure the firms average taxable income response to the exemption threshold.<sup>15</sup>

<sup>14</sup>As [Bertanha, McCallum, and Seeger \(2023\)](#) emphasize, classical bunching estimators depend critically on assumptions about the counterfactual (non-bunching) density outside the excluded region. Because our pre-reform taxable income data are missing or under-represented on the left side of the threshold, extrapolating the density into that region would be unreliable. Hence, to avoid drawing misleading conclusions from such sensitivity, we do not apply the standard bunching estimator to pre-reform observations.

<sup>15</sup>Following the bunching literature ([Saez, 2010](#); [Chetty, Friedman, Olsen, and Pistaferri, 2011](#); [Kleven and Waseem, 2013](#); [Almunia and Lopez-Rodriguez, 2018](#)), the bunching estimator  $b$  is interpreted as the average response of bunchers. Specifically,  $X_B = \int_{\bar{x}}^{x_{ub}} \hat{f}(\bar{x}) d\bar{x} \approx \frac{1}{2}[\hat{f}(\bar{x}) + \hat{f}(x_{ub})]dx$ , where  $\hat{f}(x)$  represents the counterfactual

$$dx = \frac{\hat{X}_B}{\frac{1}{2}(\hat{f}_{\bar{x}} + \hat{f}_{x_{ub}})}$$

$\hat{f}_{\bar{x}}$  and  $\hat{f}_{x_{ub}}$  are the number firm in each bin of the counterfactual distribution at the tax notch and at the upper bound of the bunching interval, respectively. Following (Chen, Liu, Suárez Serrato, and Xu, 2021; Diamond and Persson, 2016), we adopt a data-driven procedure to jointly select the excluded region  $(x_{lb}, x_{ub})$  and polynomial degree  $p$  via K-fold cross-validation. For each candidate triple  $(p, x_{lb}, x_{ub})$ , we fit model 21 on the training fold and evaluate performance on the testing fold using a custom loss function. The loss has two components: (i)  $(|X_B/X_H| - 1)^2$ , which tests the hypothesis that the excess bunching mass ( $X_B$ ) to the left of the threshold equals the missing mass ( $X_H$ ) to the right within the exclusion window; and (ii) the sum of squared prediction errors outside the excluded region. We then select the parameter triple that minimizes this loss function across folds. We apply this procedure separately to the pooled data for the pre-reform and post-reform periods. For the yearly estimator, due to the limited number of observations in individual years, we use the same polynomial degree  $p$  and lower bound parameter  $(p, D_m)$  as selected from the pooled sample.

Table B.4 reports bunching estimates for the pooled post-reform period as well as for each year separately. On average, firms reduce reported taxable income by about 27 thousand NTD to avoid exceeding the 120 thousand NTD exemption threshold, which corresponds to roughly 22% of the threshold level. The yearly estimates are broadly stable, showing no clear dynamic pattern either immediately after the reform or in subsequent years, which suggests a persistent behavioral response over the entire post-reform period with no systematic buildup or decline over time.

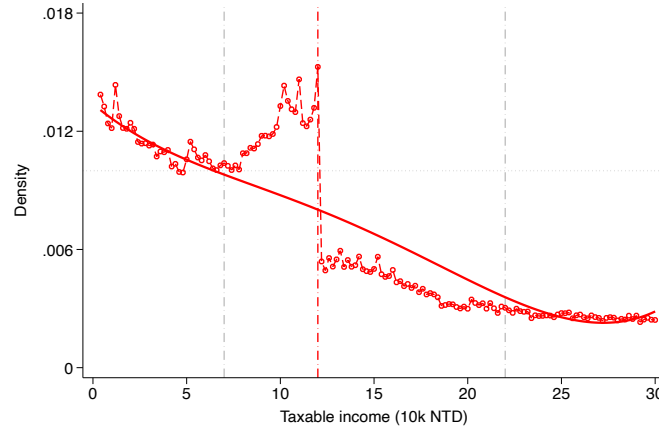
One thing worth noting is that the bunching estimation is constrained by the fraction of firms in the range  $[x_{th}, x_{ub}]$  that do not respond to the threshold, as firms with corporate taxable income above the tax threshold and below the upper bound of the exclusive region do not respond to the threshold. The fraction of firms that do not respond to exempt threshold is between 59-69%.<sup>16</sup>

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density of firms' taxable income without the tax notch. Consequently,  $dx \equiv \frac{B}{\frac{1}{2}[\hat{f}(\bar{x}) + \hat{f}(\bar{x} + dy^H)]} \approx dx$ . Once we introduce heterogeneity in firms productivity that determine their taxable income, the bunching estimator  $dx \approx E(dx)$  is interpreted as weight average of bunch's response and the lack of reaction of non-bunchers.

<sup>16</sup>Following the literature, the fraction of firms that do not respond is defined as  $a \equiv \frac{\int_{\bar{x}}^{x_{ub}} f_0(x) dx}{\int_{\bar{x}}^{x_{ub}} f(x) dx}$ .

**Figure B.1:** Estimated Counterfactual Density and the Average Bunching Response



*Notes:* This figure shows the density distribution of corporate taxable incomes after (2010-2017) with the bunching estimator reported at the right corner of each panel. The horizontal axis represents the taxable income bins. The vertical axis denotes the firms' density of corporate taxable income between 0 and 300,000 NTD. The vertical red indicates the CIT exempt threshold before and after the reform. The vertical dashed line represents the lower bound and upper bound of the exclusive region.

**Table B.4:** Bunching Estimator by Year

|        | $dx$     | SD    | $x_{lb}$ | $x_{ub}$ | $a$  | $X_B/X_H$ | N       |
|--------|----------|-------|----------|----------|------|-----------|---------|
| Pooled | 27.39*** | -1.16 | 80       | 246      | 0.76 | -1.09     | 327,289 |
| 2010   | 24.85*** | -3.32 | 80       | 268      | 0.80 | -0.99     | 40,226  |
| 2011   | 26.99*** | -2.38 | 80       | 256      | 0.77 | -1.05     | 40,222  |
| 2012   | 29.02*** | -2.04 | 80       | 252      | 0.75 | -1.04     | 38,678  |
| 2013   | 27.53*** | -1.41 | 80       | 240      | 0.74 | -1.03     | 36,393  |
| 2014   | 32.27*** | -3.77 | 80       | 262      | 0.74 | -1.02     | 36,006  |
| 2015   | 26.23*** | -1.47 | 80       | 240      | 0.75 | -0.98     | 34,531  |
| 2016   | 26.76*** | -1.64 | 80       | 250      | 0.77 | -1.04     | 33,929  |
| 2017   | 27.78*** | -1.97 | 80       | 254      | 0.76 | -0.96     | 33,472  |
| 2018   | 25.73*** | -1.38 | 80       | 234      | 0.75 | -1.03     | 33,832  |

*Notes:* Bootstrapped standard errors are shown below each bunching estimator.  $X_B$  is the number of firms difference between observed distribution and estimated distribution in the bunching region  $[x_{lb}, \bar{x}]$ ;  $X_H$  is the number of firms difference between observed distribution and estimated distribution in the range  $[\bar{x}, x_{ub}]$ ;  $X_B/X_H$  is their ratio.  $x_{ub}$  and  $x_{lb}$  are the bounds of the excluded region.  $a$  is the fraction of firms that do not respond to the exempt threshold in  $[\bar{x}, x_{ub}]$ .  $N$  is the number of observations included in the estimation (firms with corporate taxable incomes between 0 and 300 thousand NTD).

## C Alternative Empirical Strategy

We exploit a difference-in-differences strategy to examine how tax notches shapes firms' labor and intermediate input decisions. Our baseline specification is presented below.

$$y_{ijt} = \beta \text{Treated}_j \times \text{Post}_t + \lambda_{jt} + \eta_j + \epsilon_{ijt}. \quad (22)$$

We use  $y_{ijt}$  to denote the outcome variable for firm  $i$  in industry  $j$  and year  $t$ , including average input cost shares in gross revenue. The dummy variable  $\text{Treated}_j$  equals one for firms whose taxable incomes were below 50 thousand NTD in 2009 (i.e., the treatment group was subject to the exemption threshold) and zero for firms with taxable income between 50 thousand and 80 thousand NTD in 2009, which serve as a control group. The indicator  $\text{Post}_t$  takes the value of one for years after the reform (2010 and after) and zero otherwise. The coefficient  $\beta$  on the interaction term  $\text{Treated}_j \times \text{Post}_t$  thus captures the average treatment effect of lifting the corporate exemption threshold on firms' labor salary share and intermediate input share.

We control for industry-by-year fixed effects,  $\lambda_{jt}$ , which absorb common macroeconomic shocks that vary across industries or over time, such as aggregate demand fluctuations, inflation, and labor market conditions. Moreover, we include firm fixed effects,  $\eta_j$ , to control for time-invariant firm characteristics such as managerial quality or production technology. Together, these controls mitigate potential confounding factors such as the lingering effects of the 2008 global financial crisis and subsequent recovery to ensure that the estimated  $\beta$  reflects the causal impact of the 2010 corporate income tax reform.

Our key identification assumption is that conditional on industry-by-year fixed effects and firm fixed effects, the treated and control group should follow the same trend in input shares had there not been a change in the exemption threshold. Since we rely on within-firm variation over time to identify the policy impacts, we focus on the static outcome variables such as the labor cost share and intermediate input ratios. The same empirical framework is not well suited to study firms' dynamic responses, as we cannot control for firm fixed effects when analyzing exit decisions.

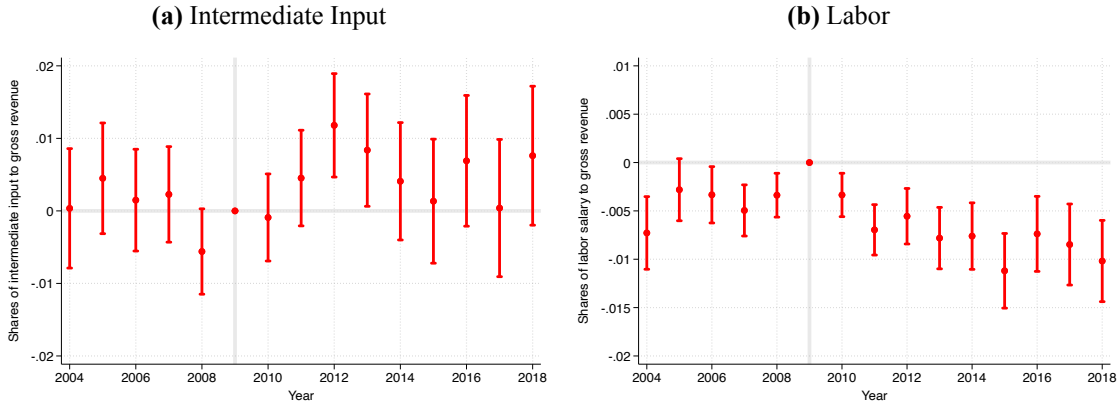
To bolster the confidence that our estimation uncovers the causal effects, we further conduct an event study to examine the parallel-trend assumption and the dynamic effects of the reform. The empirical model is shown in equation (23). Compared with equation (22), we replace  $\text{Post}_t$  with a set of dummy variables  $\mathbb{I}(t - 2010 = k)$ , which is defined as whether year  $t$  is  $k$  years after 2010. We choose 2009, one year before the reform, as the baseline year. Therefore,  $\beta_k$  measures how the outcome changes compared to 2009 due to the lift of the exemption threshold.

$$y_{ijt} = \sum_{k=-3, k \neq -1}^{k=4} \beta_k \cdot \text{Treated}_j \times \mathbb{I}(t - 2010 = k) + \lambda_{jt} + \eta_j + \epsilon_{ijt}. \quad (23)$$

Table C.1 presents the difference-in-differences estimates. We find that the removal of the 50 thousand NTD exemption threshold leads to an increase in the intermediate input share of revenue and a decrease in the labor salary share. These results suggest that firms engage in bunching by reducing actual production while over-reporting labor expenses.

Figure C.1 displays the estimated  $\beta_k$  coefficients from equation (23), showing the dynamic effect of lifting the exemption threshold on intermediate input share in panel (a) and labor cost share in panel (b). Two key insights emerge from this analysis. First, labor input adjusts immediately, whereas intermediate inputs respond with a lag. This pattern is consistent with the view that reported labor costs can be more easily manipulated for reporting purposes, while intermediate inputs reflect real production decisions that require more time to change. Second, both outcomes display parallel pre-trends prior to the reform, supporting the validity of the parallel trends assumption in our empirical design.

**Figure C.1: The Effect of Tax Notches on Input Cost Shares: Event Study**



*Notes:* These figures display the estimated coefficients  $\beta_k$  from the dynamic difference-in-difference equation (23).  $\beta_k$  is the coefficient of interest, which measures the firms' intermediate inputs and labor salary share relative to the reference year. The solid point denotes the point estimates. The line indicates the 95% confidence interval. The horizontal axis represents the number of years after the 2010 corporate income tax reform.

**Table C.1:** The Effect of Tax Notches on Input Cost Shares: DID

|                                          | (1)                  | (2)                             | (3)                  | (4)                  |
|------------------------------------------|----------------------|---------------------------------|----------------------|----------------------|
| <i>Panel A: Intermediate input share</i> |                      | <u>Intermediate input share</u> |                      |                      |
| Treatment                                | 0.014***<br>(0.003)  | 0.013***<br>(0.003)             | 0.014***<br>(0.003)  | 0.005*<br>(0.002)    |
| # of Observations                        | 160,066              | 160,066                         | 150,734              | 148,329              |
| <i>Panel B: Labor salary share</i>       |                      | <u>Labor salary share</u>       |                      |                      |
| Treatment                                | -0.008***<br>(0.002) | -0.008***<br>(0.002)            | -0.009***<br>(0.002) | -0.004***<br>(0.001) |
| # of Observations                        | 159,454              | 159,454                         | 151,050              | 148,586              |
| Basic DID controls                       | ✓                    | ✓                               | ✓                    | ✓                    |
| Year fixed effect                        |                      | ✓                               |                      |                      |
| Industry-year linear trends              |                      |                                 | ✓                    | ✓                    |
| Firm fixed effects                       |                      |                                 |                      | ✓                    |

*Notes:* This table reports the estimation results using empirical model in equation (22). The dependent variable is firms' intermediate input share (Panel A) and labor salary share (Panel B). Column (1) follows the basic DID setup, with variables  $Post_t$  and  $Treated_i$ . Column (2) additionally includes year fixed effects. Column (3) further considers industry-year linear-time trends. Column (4) includes firm fixed effects. Standard errors are clustered at the firm level in parentheses. \*p<0.10; \*\*p<0.05; \*\*\*p<0.01.